

Maintenance and Troubleshooting of Electric Motors

CHAPTER 1 - Motor Maintenance

SCHEDULED ROUTINE CARE

Introduction

The key to minimizing motor problems is scheduled routine inspection and service. The frequency of routine service varies widely between applications.

Including the motors in the maintenance schedule for the driven machine or general plant equipment is usually sufficient. A motor may require additional or more frequent attention if a breakdown would cause health or safety problems, severe loss of production, damage to expensive equipment or other serious losses.

Written records indicating date, items inspected, service performed and motor condition are important to an effective routine maintenance program. From such records, specific problems in each application can be identified and solved routinely to avoid breakdowns and production losses.

The routine inspection and servicing can generally be done without disconnecting or disassembling the motor. It involves the following factors:

Dirt and Corrosion

1. Wipe, brush, vacuum or blow accumulated dirt from the frame and air passages of the motor. Dirty motors run hot when thick dirt insulates the frame and clogged passages reduce cooling air flow. Heat reduces insulation life and eventually causes motor failure.
2. Feel for air being discharged from the cooling air ports. If the flow is weak or unsteady, internal air passages are probably clogged. Remove the motor from service and clean.
3. Check for signs of corrosion. Serious corrosion may indicate internal deterioration and/or a need for external repainting. Schedule the removal of the motor from service for complete inspection and possible rebuilding.
4. In wet or corrosive environments, open the conduit box and check for deteriorating insulation or corroded terminals. Repair as needed.

Lubrication

Lubricate the bearings only when scheduled or if they are noisy or running hot. Do NOT over-lubricate. Excessive grease and oil creates dirt and can damage bearings. See "[Bearing Lubrication](#)" for more details.

Heat, Noise and Vibration

Feel the motor frame and bearings for excessive heat or vibration. Listen for abnormal noise. All indicate a possible system failure. Promptly identify and eliminate the source of the heat, noise or vibration. See "[Heat, Noise and Vibration](#)" for details.

Winding Insulation

When records indicate a tendency toward periodic winding failures in the application, check the condition of the insulation with an insulation resistance test. See "[Testing Windings](#)" for details. Such testing is especially important for motors operated in wet or corrosive atmospheres or in high ambient temperatures.

Brushes and Commutators (DC Motors)

1. Observe the brushes while the motor is running. The brushes must ride on the commutator smoothly with little or no sparking and no brush noise (chatter).
2. Stop the motor. Be certain that:
 - a. The brushes move freely in the holder and the spring tension on each brush is about equal.
 - b. Every brush has a polished surface over the entire working face indicating good seating.
 - c. The commutator is clean, smooth and has a polished brown surface where the brushes ride.

NOTE: Always put each brush back into its original holder. Interchanging brushes decreases commutation ability.

 - d. There is no grooving of the commutator (small grooves around the circumference of the commutator). If there is grooving, remove the motor from service immediately as this is a symptomatic indication of a very serious problem.
3. Replace the brushes if there is any chance they will not last until the next inspection date.
4. If accumulating, clean foreign material from the grooves between the commutator bars and from the brush holders and posts.
5. Brush sparking, chatter, excessive wear or chipping, and a dirty or rough commutator indicate motor problems requiring prompt service. See "[Brush and Commutator Care](#)" for details.



Figure 1. Typical DC Motor Brushes And Commutator

Brushes and Collector Rings (Synchronous Motors)

1. Black spots on the collector rings must be removed by rubbing lightly with fine sandpaper. If not removed, these spots cause pitting that requires regrinding the rings.

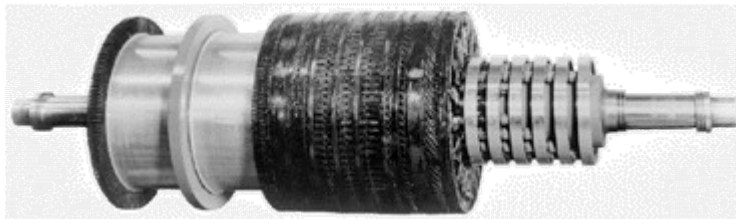


Figure 2. Rotary Converter Armature Showing Commutator And Slip Rings.

2. An imprint of the brush, signs of arcing or uneven wear indicate the need to remove the motor from service and repair or replace the rings.
3. Check the collector ring brushes as described under "[Brushes and Commutators](#)". They do not, however, wear as rapidly as commutator brushes.

BEARING LUBRICATION

Introduction

Modern motor designs usually provide a generous supply of lubricant in tight bearing housings. Lubrication on a scheduled basis, in conformance with the manufacturer's recommendations, provides optimum bearing life.

Thoroughly clean the lubrication equipment and fittings before lubricating. Dirt introduced into the bearings during lubrication probably causes more bearing failures than the lack of lubrication.

Too much grease can overpack bearings and cause them to run hot, shortening their life. Excessive lubricant can find its way inside the motor where it collects dirt and causes insulation deterioration.

Many small motors are built with permanently lubricated bearings. They cannot and should not be lubricated.

Oiling Sleeve Bearings

As a general rule, fractional horsepower motors with a wick lubrication system should be oiled every 2000 hours of operation or at least annually. Dirty, wet or corrosive locations or heavy loading may require oiling at three-month intervals or more often. Roughly 30 drops of oil for a 3-inch diameter frame to 100 drops for a 9-inch diameter frame is sufficient. Use a 150 SUS viscosity turbine oil or SAE 10 automotive oil.

Some larger motors are equipped with oil reservoirs and usually a sight gage to check proper level.

([Figure 3](#)) As long as the oil is clean and light in color, the only requirement is to fill the cavity to the proper level with the oil recommended by the manufacturer. Do not overfill the cavity. If the oil is discolored, dirty or contains water, remove the drain plug. Flush the bearing with fresh oil until it comes out clean. Coat the plug threads with a sealing compound, replace the plug and fill the cavity to the proper level.

When motors are disassembled, wash the housing with a solvent. Discard used felt packing. Replace badly worn bearings. Coat the shaft and bearing surfaces with oil and reassemble.

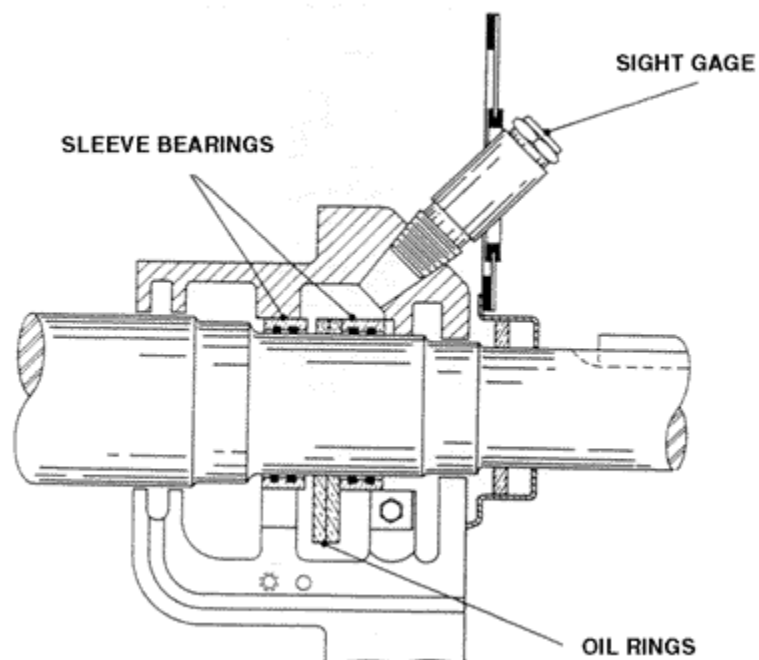


Figure 3. Cross Section of the Bearing System of a Large Motor

Greasing Ball and Roller Bearings

Practically all Reliance ball bearing motors in current production are equipped with the exclusive PLS/Positive Lubrication System. PLS is a patented open-bearing system that provides long, reliable bearing and motor life regardless of mounting position. Its special internal passages uniformly distribute new grease pumped into the housing during regreasing through the open bearings and forces old grease out through the drain hole. The close running tolerance between shaft and inner bearing cap minimizes entry of contaminants into the housing and grease migration into the motor. The unique V-groove outer slinger seals the opening between the shaft and end bracket while the motor is running or is at rest yet allows relief of grease along the shaft if the drain hole is plugged. ([Figure 4](#))

The frequency of routine greasing increases with motor size and severity of the application as indicated in Table 1. Actual schedules must be selected by the user for the specific conditions.

During scheduled greasing, remove both the inlet and drain plugs. Pump grease into the housing using a standard grease gun and light pressure until clean grease comes out of the drain hole.

If the bearings are hot or noisy even after correction of bearing overloads (see "Troubleshooting") remove the motor from service. Wash the housing and bearings with a good solvent. Replace bearings that show signs of damage or wear. Repack the bearings, assemble the motor and fill the grease cavity.

Whenever motors are disassembled for service, check the bearing housing. Wipe out any old grease. If there are any signs of grease contamination or breakdown, clean and repack the bearing system as described in the preceding paragraph.

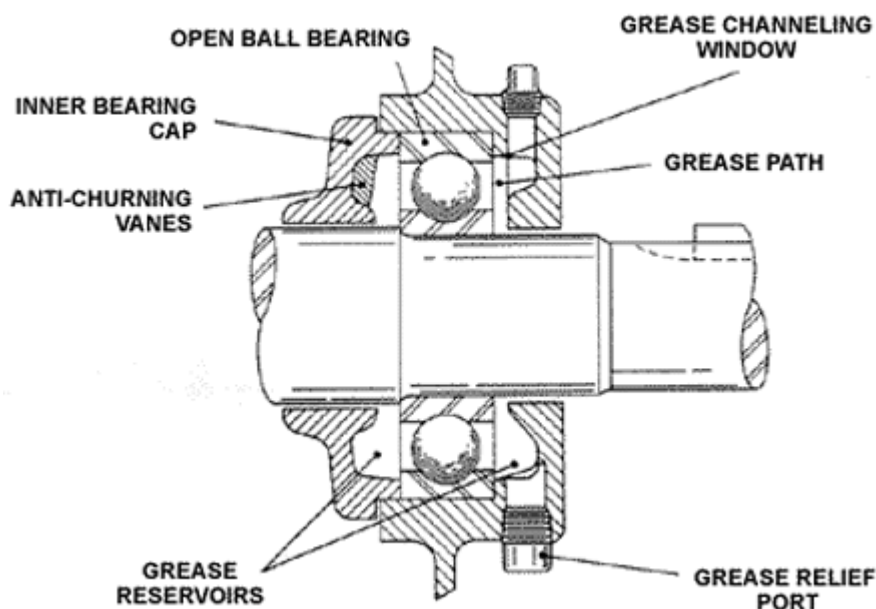


Figure 4. Cross Section of PLS Bearing System (Positive Lubrication System)

HEAT, NOISE AND VIBRATION

Heat

Excessive heat is both a cause of motor failure and a sign of other motor problems.

The primary damage caused by excess heat is to increase the aging rate of the insulation. Heat beyond the insulation's rating shortens winding life. After overheating, a motor may run satisfactorily but its useful life will be shorter. For maximum motor life, the cause of overheating should be identified and eliminated.

As indicated in the Troubleshooting Sections, overheating results from a variety of different motor problems. They can be grouped as follows:

1. **WRONG MOTOR:** It may be too small or have the wrong starting torque characteristics for the load. This may be the result of poor initial selection or changes in the load requirements.
2. **POOR COOLING:** Accumulated dirt or poor motor location may prevent the free flow of cooling air around the motor. In other cases, the motor may draw heated air from another source. Internal dirt or damage can prevent proper air flow through all sections of the motor. Dirt on the frame may prevent transfer of internal heat to the cooler ambient air.
3. **OVERLOADED DRIVEN MACHINE:** Excess loads or jams in the driven machine force the motor to supply higher torque, draw more current and overheat.

Table 1. Motor Operating Conditions

Motor Horsepower	Light Duty ⁽¹⁾	Standard Duty ⁽²⁾	Heavy Duty ⁽³⁾	Severe Duty ⁽⁴⁾
Up to 7-1/2	10	7 years	4 years	9
10 to 40	years	4 years	1-1/2	months
50 to 150	7 years	1-1/2	years	4
Over 150	4 years	years	9	months
	1 year	6 months	months	3
			3	months
			months	2
				months

1. **Light Duty:** Motors operate infrequently (1 hour/day or less) as in portable floor sanders, valves, door openers.
2. **Standard Duty:** Motors operate in normal applications (1 or 2 work shifts). Examples include air conditioning units, conveyors, refrigeration apparatus, laundry machinery, woodworking and textile machines, water pumps, machine tools, garage compressors.
3. **Heavy Duty:** Motors subjected to above normal operation and vibration (running 24 hours/day, 365 days/year). Such operations as in steel mill service, coal and mining machinery, motor-generator sets, fans, pumps.

4. *Severe Duty: Extremely harsh, dirty motor applications. Severe vibration and high ambient conditions often exist.*

4. **EXCESSIVE FRICTION:** Misalignment, poor bearings and other problems in the driven machine, power transmission system or motor increase the torque required to drive the loads, raising motor operating temperature.

5. **ELECTRICAL OVERLOADS:** An electrical failure of a winding or connection in the motor can cause other Windings or the entire motor to overheat.

Noise and Vibration

Noise indicates motor problems but ordinarily does not cause damage. Noise, however, is usually accompanied by vibration.

Vibration can cause damage in several ways. It tends to shake windings loose and mechanically damages insulation by cracking, flaking or abrading the material. Embrittlement of lead wires from excessive movement and brush sparking at commutators or current collector rings also results from vibration. Finally, vibration can speed bearing failure by causing balls to "brinnell," sleeve bearings to be pounded out of shape or the housings to loosen in the shells.

Whenever noise or vibration is found in an operating motor, the source should be quickly isolated and corrected. What seems to be an obvious source of the noise or vibration may be a symptom of a hidden problem. Therefore, a thorough investigation is often required.

Noise and vibrations can be caused by a misaligned motor shaft or can be transmitted to the motor from the driven machine or power transmission system. They can also be the result of either electrical or mechanical unbalance in the motor.

After checking the motor shaft alignment, disconnect the motor from the driven load. If the motor then operates smoothly, look for the source of noise or vibration in the driven equipment.

If the disconnected motor still vibrates, remove power from the motor. If the vibration stops, look for an electrical unbalance. If it continues as the motor coasts without power, look for a mechanical unbalance.

Electrical unbalance occurs when the magnetic attraction between stator and rotor is uneven around the periphery of the motor. This causes the shaft to deflect as it rotates creating a mechanical unbalance. Electrical unbalance usually indicates an electrical failure such as an open stator or rotor winding, an open bar or ring in squirrel cage motors or shorted field coils in synchronous motors. An uneven air gap, usually from badly worn sleeve bearings, also produces electrical unbalance.

The chief causes of mechanical unbalance include a distorted mounting, bent shaft, poorly balanced rotor, loose parts on the rotor or bad bearings. Noise can also come from the fan hitting the frame, shroud, or foreign objects inside the shroud. If the

bearings are bad, as indicated by excessive bearing noise, determine why the bearings failed. (See Troubleshooting Problems D and L.)

Brush chatter is a motor noise that can be caused by vibration or other problems unrelated to vibration. See Troubleshooting Problem M for details.

WINDINGS

Care of Windings and Insulation

Except for expensive, high horsepower motors, routine inspections generally do not involve opening the motor to inspect the windings. Therefore, long motor life requires selection of the proper enclosure to protect the windings from excessive dirt, abrasives, moisture, oil and chemicals.

When the need is indicated by severe operating conditions or a history of winding failures, routine testing can identify deteriorating insulation. Such motors can be removed from service and repaired before unexpected failures stop production. See "[Testing Windings](#)".

Whenever a motor is opened for repair, service the windings as follows:

1. Accumulated dirt prevents proper cooling and may absorb moisture and other contaminants that damage the insulation. Vacuum the dirt from the windings and internal air passages. Do not use high pressure air because this can damage windings by driving the dirt into the insulation.
2. Abrasive dust drawn through the motor can abrade coil noses, removing insulation. If such abrasion is found, the winding should be revarnished or replaced.
3. Moisture reduces the dielectric strength of insulation which results in shorts. If the inside of the motor is damp, dry the motor per information in "[Cleaning and Drying Windings](#)".
4. Wipe any oil and grease from inside the motor. Use care with solvents that can attack the insulation.
5. If the insulation appears brittle, overheated or cracked, the motor should be revarnished or, with severe conditions, rewind.
6. Loose coils and leads can move with changing magnetic fields or vibration, causing the insulation to wear, crack or fray. Revarnishing and retying leads may correct minor problems. If the loose coil situation is severe, the motor must be rewind.
7. Check the lead-to-coil connections for signs of overheating or corrosion. These connections are often exposed on large motors but taped on small motors. Repair as needed.
8. Check wound rotor windings as described for stator windings. Because rotor windings must withstand centrifugal forces, tightness is even more important. In addition, check for loose pole pieces or other loose parts that create unbalance problems.

9. The cast rotor rods and end rings of squirrel cage motors rarely need attention. However, open or broken rods create electrical unbalance that increases with the number of rods broken. An open end ring causes severe vibration and noise.

Testing Windings

Routine field testing of windings can identify deteriorating insulation permitting scheduled repair or replacement of the motor before its failure disrupts operations. Such testing is good practice especially for applications with severe operating conditions or a history of winding failures and for expensive, high horsepower motors and locations where failures can cause health and safety problems or high economic loss.

The easiest field test that prevents the most failures is the ground-insulation, or "megger," test. It applies DC voltage, usually 500 or 1000 volts, to the motor and measures the resistance of the insulation.

NEMA standards require a minimum resistance to ground at 40 degrees C ambient of 1 megohm per kv of rating plus 1 megohm. Medium size motors in good condition will generally have megohmmeter readings in excess of 50 megohms. Low readings may indicate a seriously reduced insulation condition caused by contamination from moisture, oil or conductive dirt or deterioration from age or excessive heat.

One megger reading for a motor means little. A curve recording resistance, with the motor cold and hot, and date indicates the rate of deterioration. This curve provides the information needed to decide if the motor can be safely left in service until the next scheduled inspection time.

The megger test indicates ground insulation condition. It does not, however, measure turn-to-turn insulation condition and may not pick up localized weaknesses. Moreover, operating voltage peaks may stress the insulation more severely than megger voltage. For example, the DC output of a 500-volt megger is below the normal 625-volt peak each half cycle of an AC motor operating on a 440-volt system. Experience and conditions may indicate the need for additional routine testing.

A test used to prove existence of a safety margin above operating voltage is the AC high potential ground test. It applies a high AC voltage (typically, 65% of a voltage times twice the operating voltage plus 1000 volts) between windings and frame.

Although this test does detect poor insulation condition, the high voltage can arc to ground, burning insulation and frame, and can also actually cause failure during the test. It should never be applied to a motor with a low megger reading.

DC rather than AC high potential tests are becoming popular because the test equipment is smaller and the low test current is less dangerous to people and does not create damage of its own.

Cleaning and Drying Windings

Motors which have been flooded or which have low megger readings because of contamination by moisture, oil or conductive dust should be thoroughly cleaned and dried. The methods depend upon available equipment.

A hot water hose and detergents are commonly used to remove dirt, oil, dust or salt concentrations from rotors, stators and connection boxes. After cleaning, the windings must be dried, commonly in a forced-draft oven. Time to obtain acceptable megger readings varies from a couple hours to a few days.

BRUSH AND COMMUTATOR CARE

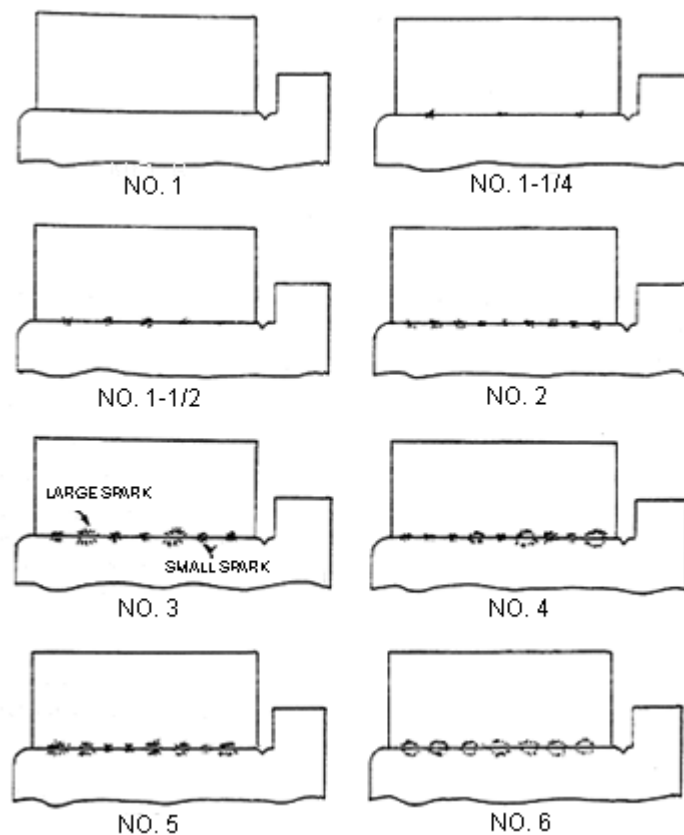
Some maintenance people with many relatively trouble-free AC squirrel cage motors forget that brushes and commutators require more frequent routine inspection and service. The result can be unnecessary failures between scheduled maintenance.

As indicated in Troubleshooting Problem M on Page 27, many factors are involved in brush and commutator problems. All generally involve brush sparking usually accompanied by chatter and often excessive wear or chipping. Sparking may result from poor commutator conditions or it may cause them.

The degree of sparking should be determined by careful visual inspection. The illustrations shown in [Figure 5](#) are a useful guide. It is very important that you gauge the degree number as accurately as possible. The solution to the problem may well depend upon the accuracy of your answer since many motor, load, environmental and application conditions can cause sparking.

It is also imperative that a remedy be determined as quickly as possible. Sparking generally feeds upon itself and becomes worse with time until serious damage results.

Some of the causes are obvious and some are not. Some are constant and others intermittent. Therefore, eliminating brush sparking, especially when it is a chronic or recurring problem, requires a thorough review of the motor and operating conditions. Always recheck for sparking after correcting one problem to see that it solved the total problem. Also remember that, after grinding the commutator and properly reseating the brushes, sparking will occur until the polished, brown surface reforms on the commutator.



NOTE:

Small sparks are yellow in color, and the large sparks are white in color. The white sparks, or blue-white sparks, are most detrimental to commutation (both brush and commutator).

Figure 5. Degrees of Generator and Motor Sparking

First consider external conditions that affect commutation. Frequent motor overloads, vibration and high humidity cause sparking. Extremely low humidity allows brushes to wear through the needed polished brown commutator surface film. Oil, paint, acid and other chemical vapors in the atmosphere contaminate brushes and the commutator surface.

Look for obvious brush and brush holder deficiencies:

1. Be sure brushes are properly seated, move freely in the holders and are not too short.
2. The brush spring pressure must be equal on all brushes.
3. Be sure spring pressure is not too light or too high. Large motors with adjustable springs should be set at about 3 to 4 pounds per square inch of brush surface in contact with the commutators.
4. Remove dust that can cause a short between brush holders and frame.
5. Check lead connections to the brush holders. Loose connections cause overheating.

Look for obvious commutator problems:

1. Any condition other than a polished, brown surface under the brushes indicates a problem. Severe sparking causes a rough blackened surface. An oil film, paint spray, chemical contamination and other abnormal conditions can cause a blackened or discolored surface and sparking. Streaking or grooving under only some brushes or flat and burned spots can result from a load mismatch and cause motor electrical problems. Grooved commutators should be removed from service. A brassy appearance shows excessive wear on the surface resulting from low humidity or wrong brush grade.
2. High mica or high or low commutator bars make the brushes jump, causing sparking.
3. Carbon dust, copper foil or other conductive dust in the slots between commutator bars causes shorting and sometimes sparking between bars.

If correcting any obvious deficiencies does not eliminate sparking or noise, look to the less obvious possibilities:

1. If brushes were changed before the problem became apparent, check the grade of brushes. Weak brushes may chip. Soft, low abrasive brushes may allow a thick film to form. High friction or high abrasion brushes wear away the brown film, producing a brassy surface. If the problem appears only under one or more of the brushes, two different grades of brushes may have been installed. Generally, use only the brushes recommended by the motor manufacturer or a qualified brush expert.
2. The brush holder may have been reset improperly. If the boxes are more than 1/8" from the commutator, the brushes can jump or chip. Setting the brush holder off neutral causes sparking. Normally the brushes must be equally spaced around the commutator and must be parallel to the bars so all make contact with each bar at the same time.
3. An eccentric commutator causes sparking and may cause vibration. Normally, concentricity should be within .001" on high speed, .002" on medium speed and .004" on slow speed motors.
4. Various electrical failures in the motor windings or connections manifest themselves in sparking and poor commutation. Look for shorts or opens in the armature circuit and for grounds, shorts or opens in the field winding circuits. A weak interpole circuit or large air gap also generate brush sparking.

CHAPTER 2 - Troubleshooting AC Motors

Problem A - Motor won't start or motor accelerates too slowly

A1: Check input power to starter.
Is there power on all lines?
(Three-phase motors won't start
on one-phase.)

■ **NO** ➔

Restore power on all lines

■ **YES** ↓

A2: Check starter. Is overload
protection device opened?

■ **YES** ➔

Replace or reset device.
Does it open again when
starting?

■ **NO** ↓

■ **YES** ➔

A3: Is there power on all lines to
motor?

■ **NO** ➔

Repair starter

■ **YES** ↓

A4: Is voltage to motor more than
10% below nameplate voltage?

■ **YES** ➔

Restore proper voltage.

■ **NO** ↓

A5: Check motor terminal
connections. Are any loose or
broken?

■ **YES** ➔

Repair connections.

■ **NO** ↓

A6: May be wrong motor for
application. Is starting load too
high?

■ **YES** ➔

Install Design C or Design
D motor. Install larger
motor.

■ **NO** ↓

A7: Is driven machine jammed or
overloaded?

■ **YES** ➔

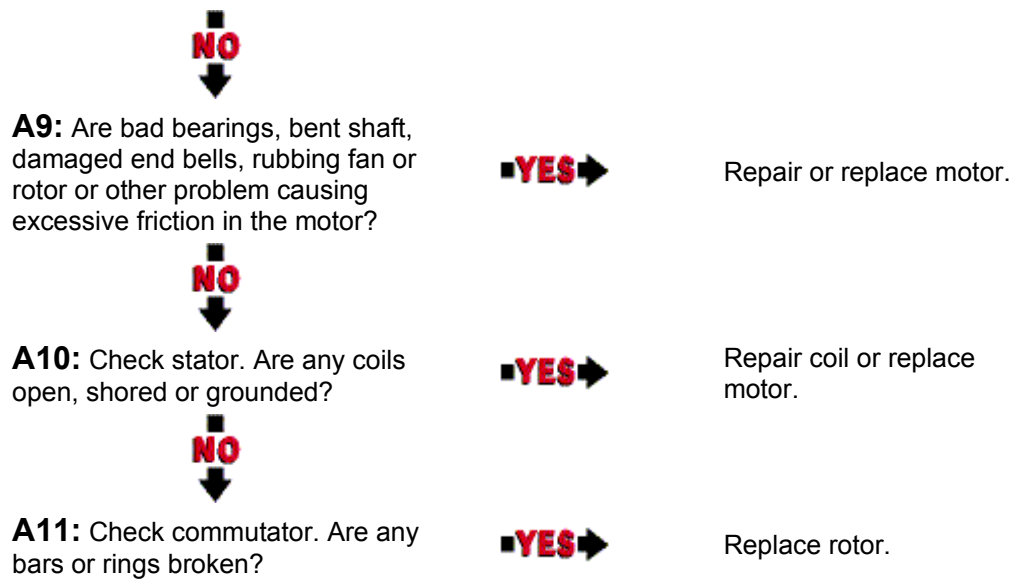
Remove jam or overload.

■ **NO** ↓

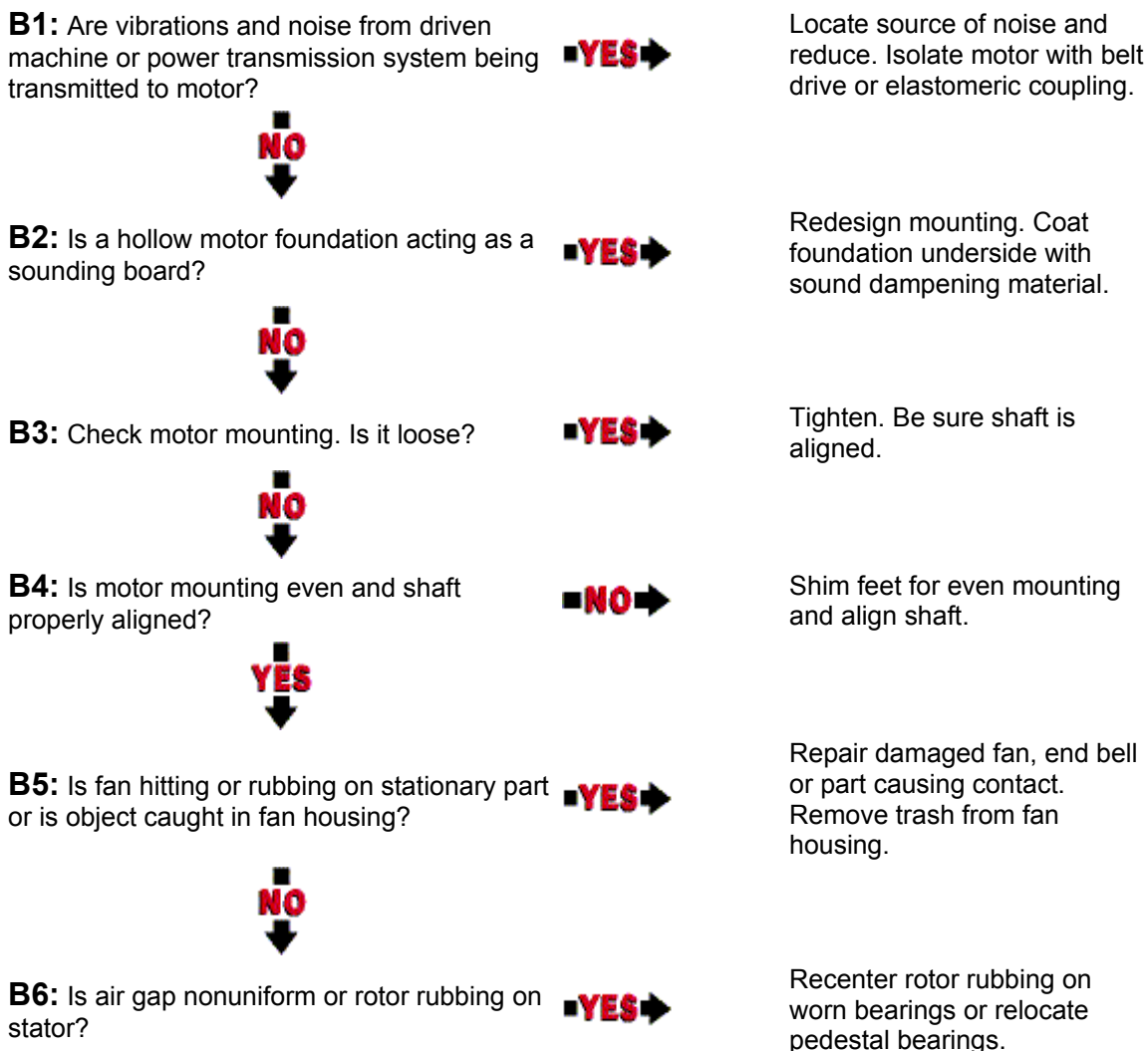
A8: Are misalignments, bad
bearings or damaged components
causing excessive friction in
driven machine or power
transmission system?

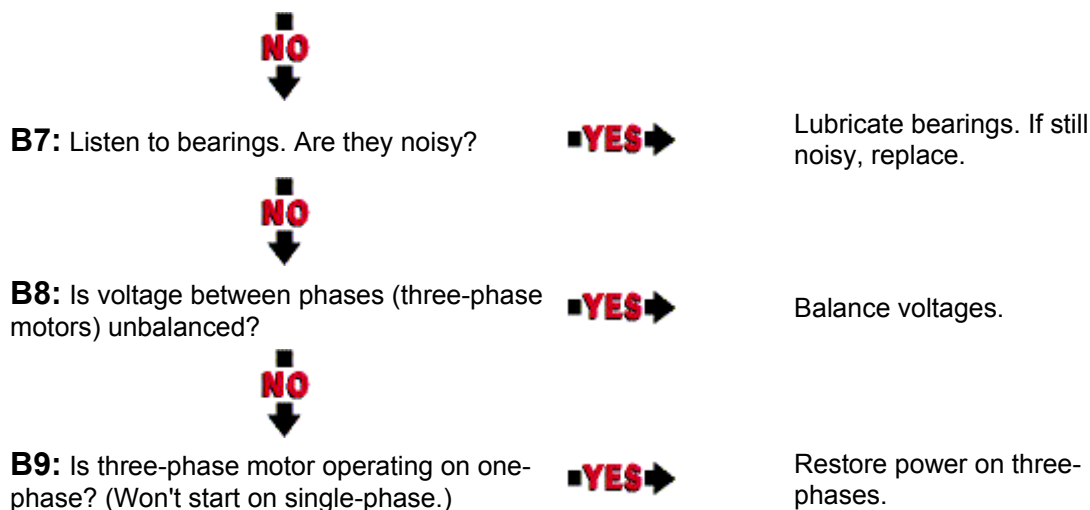
■ **YES** ➔

Repair or replace
component.

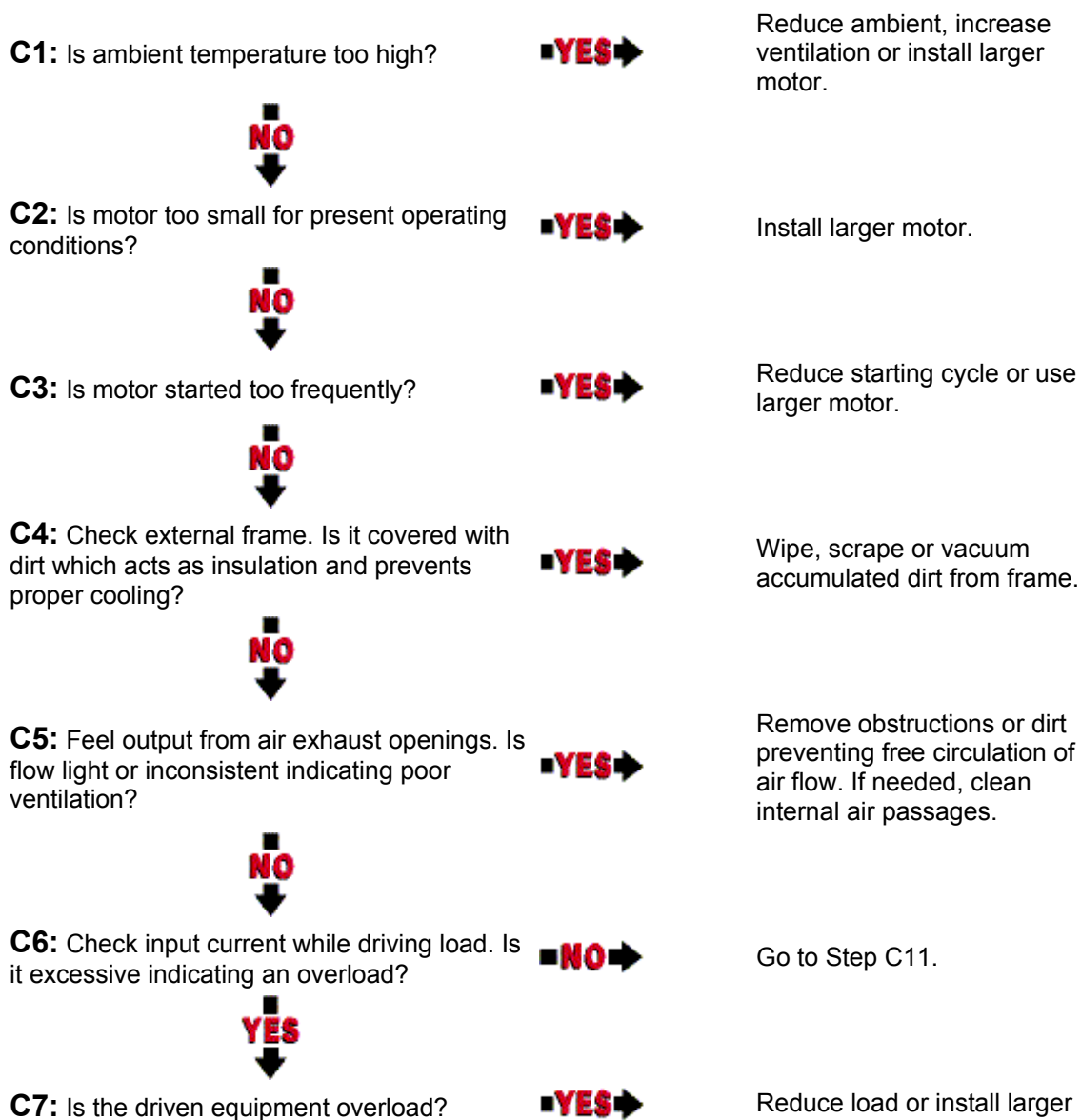


Problem B - Motor runs noisy

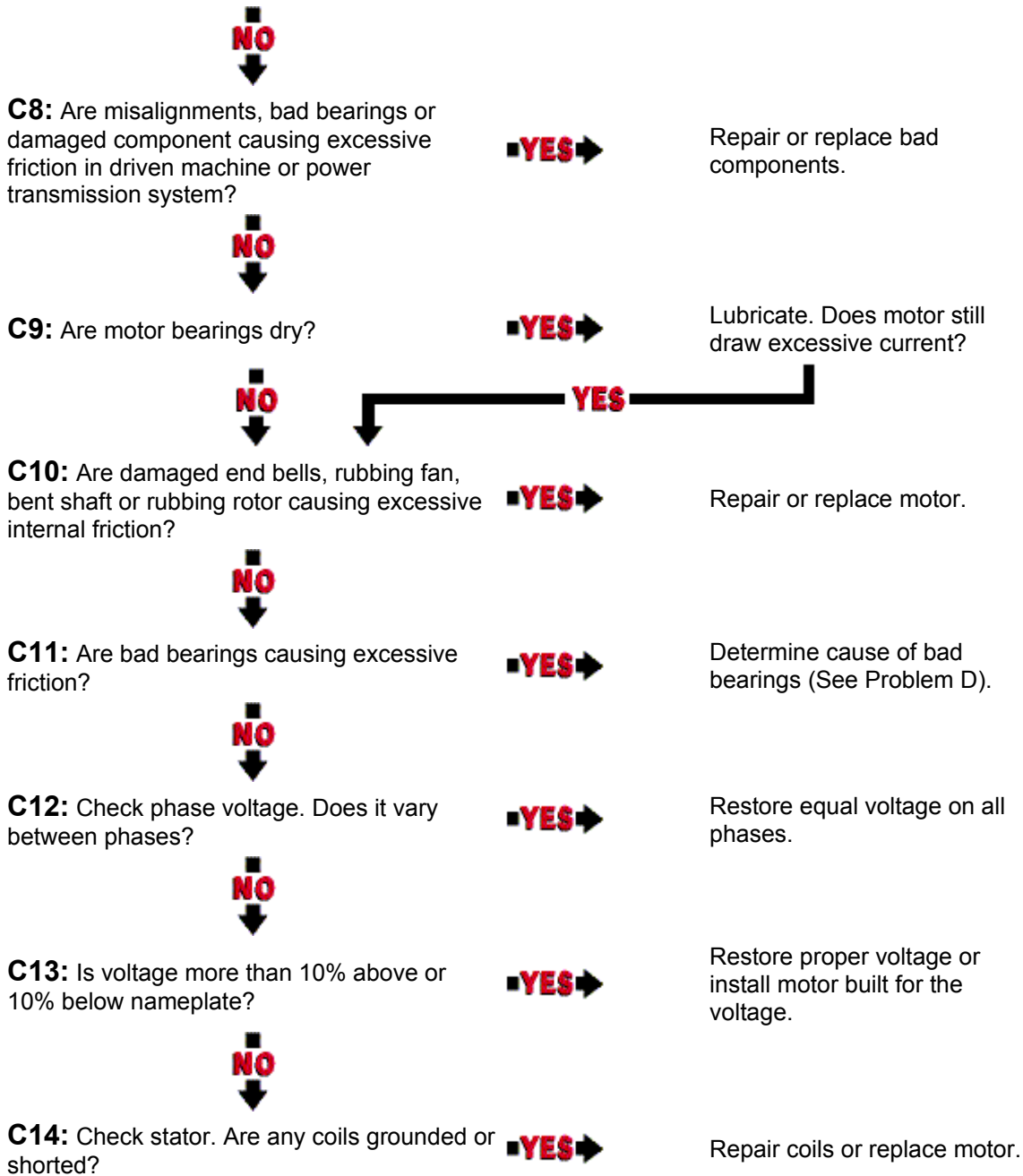




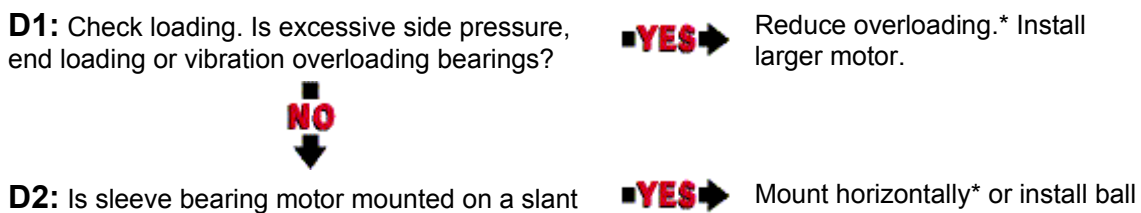
Problem C - Motor overheats



motor.



Problem D - Motor bearings run hot or noisy



causing end thrust?

bearing motor.

NO
↓

D3: Is bent or misaligned shaft overloading bearings?

■ **YES** ➔ Replace bent shaft or align shaft.*

NO
↓

D4: Is loose or damaged end bell overloading shaft?

■ **YES** ➔ Tighten or replace end bell.*

NO
↓

D5: Are bearings dry?

■ **YES** ➔ Lubricate.*

NO
↓

D6: Is bearing lubricant dirty, contaminated or of wrong grade?

■ **YES** ➔ Clean bearings and lubricate with proper grade*

NO
↓

D7: Remove end bells. Are bearings misaligned, worn or damaged?

■ **YES** ➔ Replace.

**Bearings may have been damaged. If motor still runs noisy or hot, replace bearings.*

CHAPTER 3 - Troubleshooting DC Motors

Problem E - Motor won't start

E1: Check main input power to controller. Is there power on the lines? Are contacts closed?

YES

NO

Restore input power.

E2: Check controller. Is the overload protective device open?

NO

YES

Reset or replace device. Does it open again when starting motor?

YES

E3: Check controller. Is there voltage available at output terminals?

YES

NO

Check controller for open starting resistor, broken leads and connections or other malfunctions. Repair.

E4: Set the controller for full speed. Is the voltage for field or armature circuits too low?

NO

YES

Check voltage from power source. Correct if too low. Check controller for malfunction. Repair.

E5: Check for weak or non-existent field. Is motor field open? Has one field coil shorted?

NO
↓

E6: Check for open armature circuit. Is voltage at motor armature terminals zero when starting?

NO
↓

E7: Is driven machine jammed or overloaded?

NO
↓

E8: Are misalignments, bad bearings or worn components causing excessive friction in driven machine or power transmission system?

NO
↓

E9: Are bad bearings, bent shaft, rubbing fan or rotor, damaged end bells, or other mechanical problems causing excessive friction in motor?

■ **YES** →

Repair broken leads or connections. Rewind or replace open or shorted coil.

■ **YES** →

Repair damaged armature circuit.

■ **YES** →

Remove jam or overload or install larger motor.

■ **YES** →

Correct misalignment or repair or replace worn component.

■ **YES** →

Repair or replace damaged motor components or install new motor.

Problem F - Motor starts but stops and reverses direction

F1: Check polarity

■ **YES** →

Determine

of power source. Did it reverse?



F2: Shunt and series field may be bucking each other. To check and correct: Reconnect the shunt or series field to correct polarity. Connect armature for desired rotation direction. Try fields separately to determine rotation direction and connect so both give the same rotation.

why power supply reversed polarity and repair.

Problem G - Motor runs but overload protective device trips too often.

G1: Is motor too small for load? Have loading conditions changed?



Reduce load or install larger motor.



G2: Check controller. Is overload device set too low for application?



Increase overload setting. NEVER exceed safe limits specified by codes or equipment maker.



G3: Is motor overheating?



See Problem H.

Problem H - Motor overheats

H1: Is ambient temperature too high?



■ **YES** ➔

Reduce ambient, increase ventilation or install larger motor.

H2: Check external frame. Is it covered with layer of dirt which acts as insulation and prevents proper cooling?



■ **YES** ➔

Wipe, scrape or vacuum accumulated dirt from frame.

H3: Feel output from air exhaust openings. Is flow light or inconsistent indicating poor ventilation?



■ **YES** ➔

Remove obstructions or dirt preventing free of air flow. If needed, clean internal air passages.

H4: High load speed consumes extra horsepower overloading motor. Is motor operating above normal speed?



■ **YES** ➔

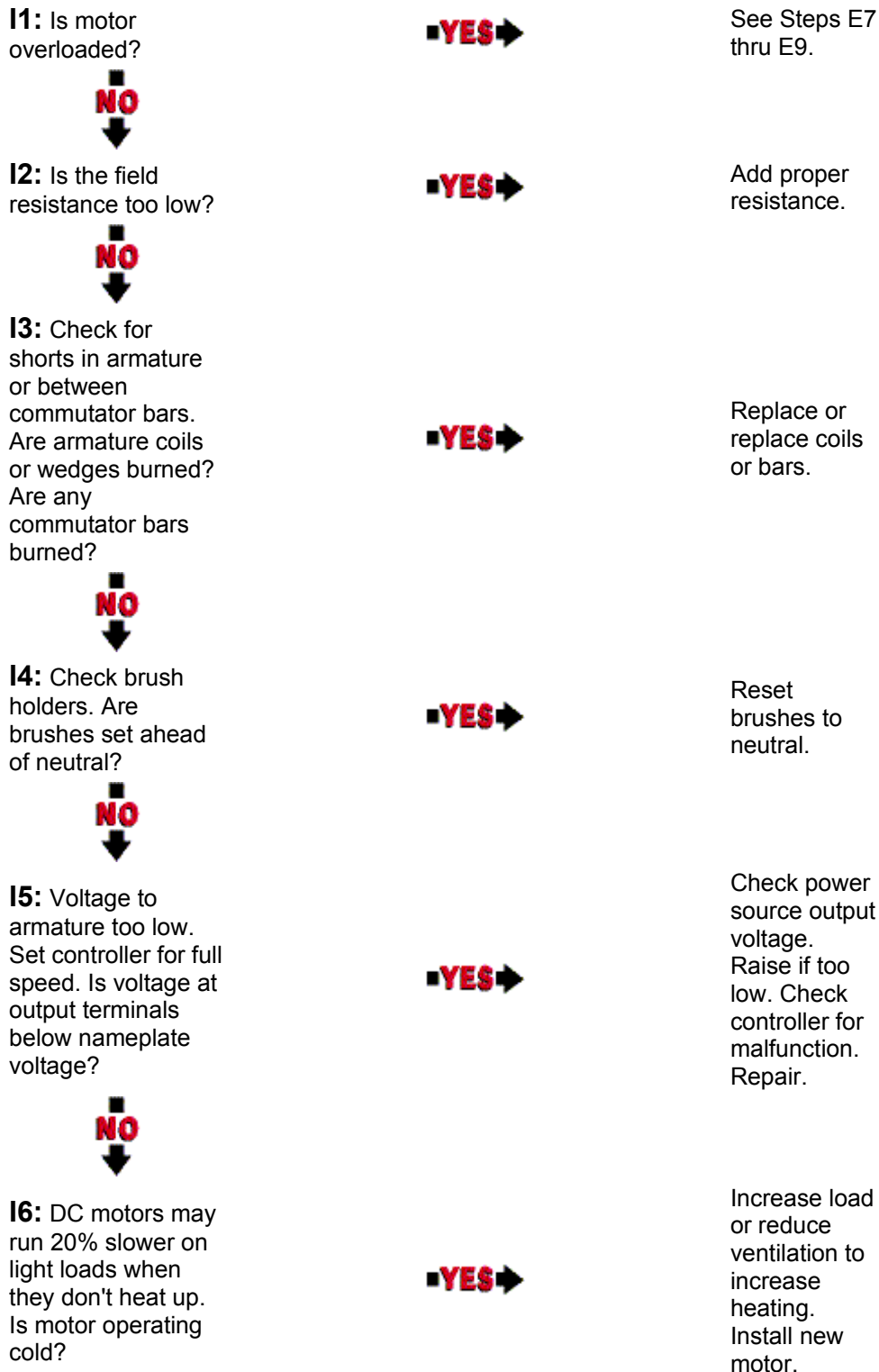
See Problem J.

H5: Check for overload.

■ **YES** ➔

See Steps E7 thru E9.

Problem I - Motor runs too slowly.



Problem J - Motor runs too fast.



motor to run fast?

NO

J2: Check for a weak field per Steps J3 through J6.

YES

J3: Are shunt or series coils reversed?

YES →

Reconnect reversed coils for proper polarity.

NO

J4: Is there excessive resistance in shunt field circuit?

YES →

Remove excessive resistance.

NO

J5: Is excessive heat causing higher resistance in shunt field circuit?

YES →

Increase ventilation or correct other cause of overheating.

NO

J6: No field causes unbalanced shunt motor to race. Is field circuit open?

YES →

Repair broken lead or connection. Replace open coil.

NO

J7: Set controller for full speed. Is voltage at output terminals of controller above nameplate voltage?

YES →

Reduce output voltage. Check controller for malfunction. Repair.

NO

J8: Check brush holders. Are brushes set behind

YES →

Reset brushes to neutral.

neutral?

Problem K - Motor runs noisy

K1: Are vibrations and noise from driven machine or power transmission system being transmitted to motor?

NO
↓

K2: Is a hollow motor foundation acting as a sounding board?

NO
↓

K3: Check motor mounting. Is it loose?

NO
↓

K4: Is motor mounting even and shaft properly aligned?

YES
↓

K5: Is fan hitting or rubbing on stationary part or is object caught in fan housing?

■ YES →

Locate source of noise and reduce. Isolate motor with belt drive or elastomeric coupling.

■ YES →

Redesign mounting. Coat foundation underside with sound dampening material.

■ YES →

Tighten. Be sure shaft is aligned.

■ NO →

Shim feet for even mounting and align shaft.

■ YES →

Repair damaged fan, end bell or part causing contact. Remove trash from fan housing.

NO
↓

K6: Is air gap nonuniform or armature rotor rubbing on pole pieces?

YES →

Tighten loose pole piece.
Recenter armature by replacing worn bearings or relocating pedestal bearings.

NO
↓

K7: Listen to bearings. Are they noisy?

YES →

Lubricate bearings. If still noisy, replace.

NO
↓

K8: Are bearings noisy or running hot?

YES →

See Problem L.

NO
↓

K9: Are the brushes developing high or low frequency chatter?

YES →

See Problem M.

Problem L - Motor bearings run hot or noisy

L1: Check loading. Is excessive side pressure, end loading or vibration overloading bearings?

YES →

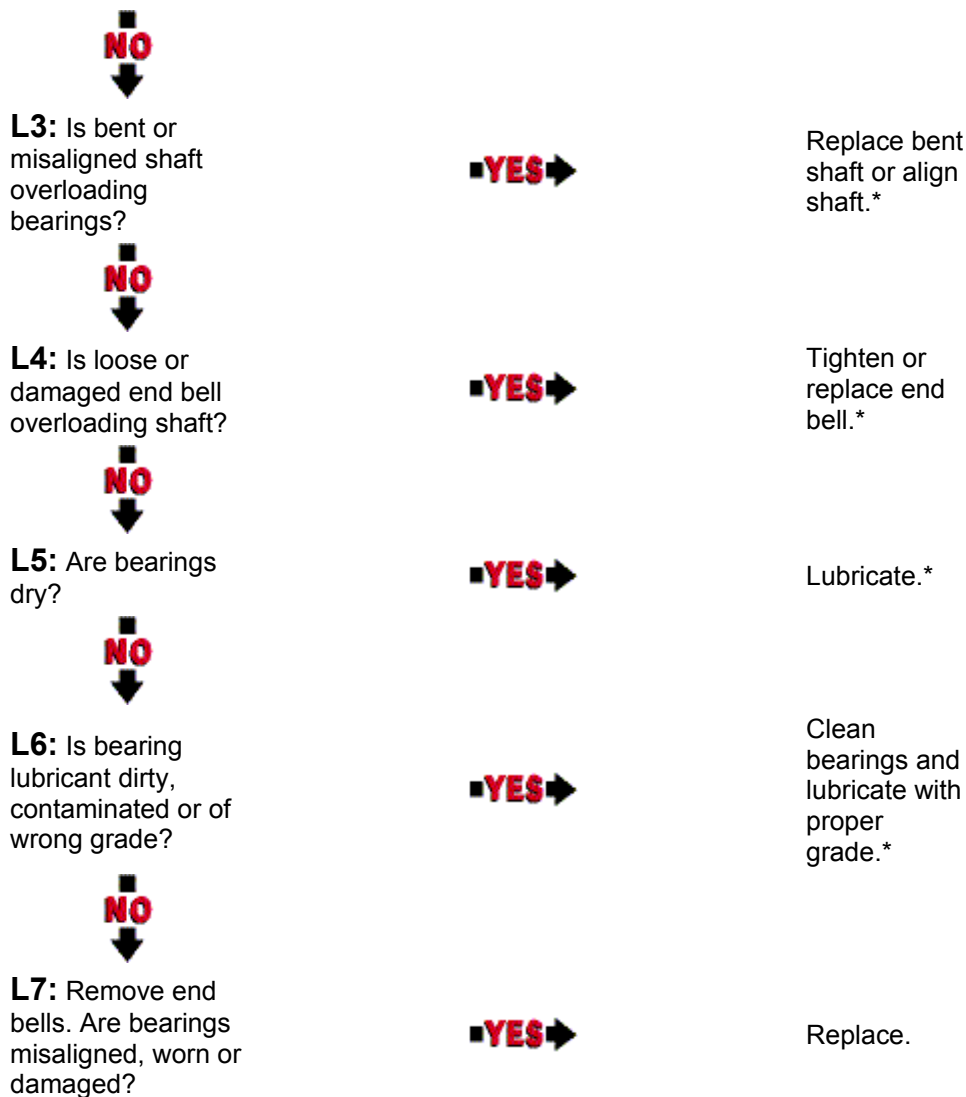
Reduce overloading.*
Install large motor.

NO
↓

L2: Is sleeve bearing motor mounted on a slant causing end thrust?

YES →

Mount horizontally*
or install ball bearing motor.



**Bearings may have been damaged. If motor still runs noisy or hot, replace bearings.*

Problem M - Brushes sparking excessively; may be accompanied by brush chatter and/or excessive wear and chipping.



present?

NO

M3: Check brushes and brush holders. Are brushes worn too short?

YES →

Replace brushes.

NO

M4: Does each brush fit commutator as indicated by polished surface over entire brush face.

NO →

Refit brushes to commutator.

YES

M5: Are brushes hanging up in holders?

YES →

Clean brushes and holders. Remove rough surfaces that cause extra friction.

NO

M6: Are brush springs broken or is spring pressure too light?

YES →

Replace spring or increase pressure. Be sure pressure is equal on all brushes.

NO

M7: Is spring pressure too high? (May also cause brush chipping)

YES →

Reduce pressure or replace with lighter spring.

NO

M8: Are brush holders set off neutral? (May also cause brush

YES →

Reset holders at neutral.

chipping)

NO
↓

M9: Are brushes set a wrong angle? (May also cause brush chipping)

■ **YES** →

Reset holders for brush angle recommended by motor manufacturer.

NO
↓

M10: Is brush holder set for more than 1/8" clearance above commutator? (May also cause brush chipping)

■ **YES** →

Reset holder for 1/8" clearance.

NO
↓

M11: Chipping brushes may also indicate wrong brush material. Are brushes too weak for duty?

■ **YES** →

Consult motor manufacturer for recommendations.

NO
↓

M12: Check commutator. Is commutator surface under brushes polished brown color?

■ **YES** →

Normal condition. Go to Step M18.

NO
↓

M13: Is commutator surface black (generally caused by sparking)?

■ **YES** →

Check for overloads, low spring tension, poorly undercut mica, loose commutator bars, etc. Correct sparking. Dress commutator.

YES
↓

M14: Is there thick film on commutator - may appear black?

■ **YES** →

Use more abrasive brushes.

NO
↓

M15: Is commutator surface bright and brassy looking?

■ **YES** →

If humidity is below 2 grams per cu. ft., increase humidity
OR
reduce spring pressure, use low friction brushes or use less abrasive brushes.

NO
↓

M16: Is commutator surface contaminated from paint spray, oil or chemical fumes? Is there excessive moisture in air?

■ **YES** →

Clean commutator and brushes and protect motor from contamination. Install motor with proper enclosure to protect commutator.

NO
↓

M17: Is commutator streaked or grooved under one or more brushes?

■ **YES** →

Be sure all brushes same grade. Replace if some are too abrasive. Check for faulty shunt connections causing unbalanced load; repair.

NO
↓

M18: Is commutator rough or eccentric?

■ **YES** →

Grind commutator round
Undercut mica

NO
↓

M19: Is mica above bar surface?

NO



M20: Are some commutator bars too high, too low or loose?

NO



M21: Are there flat or burned spots on commutator bars caused by unbalanced load in armature circuit?

NO



M22: Is conductive film carbon dust or copper flaking causing shorts between armature bars?

NO



M23: Are there any shorts or opens in armature circuits?

NO



M24: Are there any grounds, shorts or opens in the field wiring

YES

YES

YES

YES

YES

YES

Undercut mica.

Replace commutator or tighten V-ring bolts to tension recommended by manufacturer and grind commutator.

Balanced load.
Grind commutator.

Undercut mica.

Locate and repair.

Locate and repair.

circuits?

NO
↓

M25: Are connections to brush holder poor or broken?

YES →

Locate and repair.

NO
↓

M26: Is the interpole current weak or the air gap too great?

YES →

Increase interpole current or reduce gap.

Direct Current Motor Troubleshooting

”Based on the EASA TechNote “Troubleshooting DC Motors

CAUTION

ALWAYS Disconnect the power before handling any parts of the electrical equipment.

Lock out and tag out all electrical circuits.

Test for voltage before touching any components.

Check for and eliminate the danger of “stored energy” caused by raised or spring-loaded equipment.

The basic testing equipment you will need to trouble- shoot DC motors in the field includes:

Megohmmeter •

AC voltmeter

DC clamp-on ammeter

Ohmmeter

DC voltmeter

Tachometer

Find out if the failed motor was:

1. Operating successfully for a period of time before failing; or if it was
2. Installed recently.

MOTORS WITH A PREVIOUS HISTORY OF SUCCESSFUL OPERATION

If the motor has been operated successfully, problems such as incorrect hook-up or internal misconnection can be ruled out immediately.

Before proceeding,

Record pertinent motor nameplate data.

HP

RPM.

Rated voltages (Armature and field)

Rated currents (Armature and field)

Inspect the motor for any obvious defects that would prevent safe testing.

Damaged windings (Smoke, copper particles)

Loose connections (melted wire nuts, burned insulation)
Broken or missing parts (Pulleys, belts, covers, etc.)
Defective brushes or brush holders.

PROBLEM: MOTOR WILL NOT START

Check to make sure adequate AC power is available at the control unit. (Use AC volt meter)

If the main fuse is blown, DO NOT apply power to the motor until you have completed a determination of why the fuse blew.

Use the megohmmeter to measure the insulation resistance of all windings

Armature Circuit A1 – A2 or A1 – S2

Armature winding

Commutator

Brush holders

Interpoles

Series Field

(If present is probably connected with A1 and A2 leads)

Shunt Field

Any “ground” conditions must be repaired before power is applied to the motor.

If the main fuses are OK, press the start button and measure the armature and shunt field voltages at the controller using a DC voltmeter.

All output voltages must be in accordance with the motor nameplate

If rated voltage is measured, the problem is in the motor or motor wiring.

A zero or a very low reading indicates that something is wrong with the controller or control wiring.

Test and inspect controller

If no output is read from the controller, determine if the problem is in the control circuit and correct it.

Is the controller tripped?

If tripped determine cause and correct problem.

Over current (Excessive load over a period of time)

Over voltage (Overhauling type of load)
Over Heat (High ambient temperatures-Overloading)
Are the thermostats in the motor tripped (N/C contacts)?

No tach signal

Attempt a reset.

Make sure the controller is getting a start signal (N/O contacts)

Make sure there is not a STOP signal (N/C contacts)

If the controller isn't functioning by this point, it's pretty safe to say that the controller is defective.

Test and inspect motor

Inspect electrical connections to the motor.

Correct any loose or broken connections.

Check for signs of heating or “resistive connections”

Examine the brushes

Are they all making good contact on the commutator?

Are there any loose brush leads (replace any brushes that are too short or damaged)

If the motor still does not operate, disconnect the power supply from the motor

and use the ohmmeter to check the armature circuit for continuity.

An open connection in the armature circuit could be caused by:

1. Worn and hung up brushes
2. Blown brush shunts
3. Open Interpole circuit
4. Open Series Field (if so equipped)
5. Open armature (Comm. connections or winding)

PROBLEM: OVERLOAD RELAY TRIPS; OR FUSES BLOW WHEN MOTOR STARTS

A starting current that is too high causes tripping the overload relay or blowing fuses when starting.

Grounded windings.

Test all windings for ground failure using the megohmmeter. Any grounded windings must be repaired before power is applied to the motor

Mechanical problems with the motor or driven equipment.

Mechanical problems such as worn bearings or a broken pinion could cause a mechanical overload.

Determine if the problem is in the motor itself or in the driven equipment.

Uncouple the motor and turn the armature by hand. If the armature moves freely and the motor starts without tripping the overload relay or blowing fuses when uncoupled, the problem is most likely in the driven equipment and not in the motor.

Shorted armature winding

You can check the armature for shorts while the motor is uncoupled. After removing all brushes from the commutator, apply rated voltage to the shunt field and rotate the armature by hand. One or more shorted coils is indicated if the armature seems to be “bound up” or cogs as you rotate it.

If the motor will run for a short while before the overload trips or the fuse blows, shut down the motor and then feel the armature coils with your hand.

Shorted coils will feel hotter than the others because they will have had heavy circulating currents induced into the shorted turns.

Defective field winding.

A DC motor must have 100% field strength to produce its maximum torque and to keep the armature amperage within proper limits. Reduced field strength will cause high armature currents.

Test the shunt field **continuity** by measuring and recording the resistance of the shunt with an ohmmeter. A reading of *infinity* indicates an open circuit in the winding, which requires repairs to the motor.

Test the shunt field winding for **shorted turns**.

Compare your shunt field resistance measurement to the nameplate data.

The motor nameplate may tell you field resistance.

Remember, if the motor is HOT, your measured resistance will be higher than the “@ 20° C” resistance listed on the nameplate. If the resistance of the field is equal to or less than the resistance listed on the nameplate, your winding is probably shorted.

If the nameplate doesn’t tell you the correct resistance, you can calculate a good estimate using this formula:

$$\text{Field Volts/Field Amps} = \text{Field Resistance } (\Omega)$$

Note: The Field Amps on the nameplate will be correct for the motor running at it’s normal full load condition with HOT fields.

To estimate HOT resistance when you are measuring a COLD field, use this old “rule of thumb”.

$$(\Omega_{\text{cold}} \times 10) / 8 = \Omega_{\text{hot}}$$

In “Compound Wound or “Stabilized Shunt Wound” motors, test for shorts between the shunt and the series coils using the megohmmeter. Any shorts between the windings will require repairs to the motor.

Starting resistors that are shorted out prematurely. Timer or current sensing problems.

(Possible in OLD hoist controls used with Series Motors)

PROBLEM: MOTOR RUNS AT HIGHER THAN RATED RPM

Verify correct tachometer feedback.

Measure the armature and the shunt field voltages at the motor terminals to be sure they are in accordance with the nameplate data.

The speed of the motor will increase if the **armature voltage is higher** than shown on the nameplate.

The speed may also be higher if the applied **shunt field voltage is lower** than the value shown on the nameplate.

If applied voltages are in accordance with the motor rating, there is a defect in the winding.

Test for:

1. Grounds in all windings.
2. Shorts in shunt field.
4. Continuity in shunt and series coils. (High resistance indicates an open circuit.)
3. Shorts between shunt and series field (if it is a compound- wound motor).

PROBLEM: MOTOR RUNS AT LOWER THAN RATED RPM

Measure the armature and field voltage at the controller and compare it with the nameplate value.

Reduced armature voltage will decrease motor speed.

Increased field voltage will decrease motor speed.

If applied armature and field voltages are in accordance with the motor nameplate, inspect for high resistance connections in the armature circuit.

Loose connections. Check for hot spots and discolored insulation around the connections.

Contactors in controller making good contact.

Broken or “breaking” conductors due to “flexing”

Shorting contactors must be closed during operation. (Armature resistors – OLD)

If the speed of the motor varies continuously with constant voltages applied, and you have eliminated excessive resistance in the armature circuit, check for shorted armature coils.

You can check the armature for shorts while the motor is uncoupled. After removing all brushes from the commutator, apply rated voltage to the shunt field and rotate the armature by hand. One or more shorted coils is indicated if the armature seems to be “bound up” or cogs as you rotate it.

Shut down the motor and then feel the armature coils with your hand.

Shorted coils will feel hotter than the others because they will have had heavy circulating currents induced into the shorted turns.

PROBLEM: SPARKING UNDER THE BRUSHES

Sparking under the brushes is a **commutation problem**. Mechanical problems (rather than electrical ones) are usually the cause. First of all, measure the Armature current with the DC ammeter to see if the motor is overloaded. If the

armature current is OK, concentrate on mechanical problems associated with the brushes.

Inspect the Brushes

1. Make sure no brushes are missing and that all of them are properly seated on the commutator.
2. Check that all brush leads are intact and that they are securely fastened to the brush holder.
3. Check brush springs for correct pressure.
4. Make sure the brushes fit properly and move freely in their brush boxes. They should not be too tight or too loose.
5. Check brush holder mountings for looseness, which could be caused by burnt brush holder insulation (carbonization) or loose fasteners.
6. Check the brush rigging jumpers to be certain that they are tight and securely connected.
7. Inspect the brush-mounting ring for damage, and make sure it is securely locked in the neutral position.

Inspect the Commutator (mechanical focus)

1. Physical damage (for example, from rubbing).
2. Is the commutator run-out excessive.
Commutator run-out should be less than .001" when the commutator is new. The normal operating speed of the machine will dictate how much tolerance in run-out can be allowed. Generally, if the commutator run-out exceeds .005" to .010" the commutator should be turned.
3. Inspect the commutator, making certain there are no *raised* segments.
4. Make sure there are no segments with flat spots.
5. Check for *high mica*.
6. Be sure there is no foreign matter between the commutator bars
7. All of the conditions above can cause the brushes to bounce at high motor speeds. The interruption of the armature current will invariably cause sparking.
8. Severe vibrations from an unbalanced armature.
9. Run the motor uncoupled before removing it for service to see if the imbalance is in the motor or the machine.
10. Worn bearings.
11. Uneven air gaps can result if bearing clearances are out of whack. The movement of the armature can also cause the brushes to bounce.

Inspect the Commutator (electrical focus)

Inspect the commutator segments for signs of discoloration

Burned segments indicate open circuits

1. Broken coil leads behind the risers.
2. Thrown solder and loose connections at the risers.

Darkened segments can indicate shorted armature coils.

The armature can be tested as we mentioned earlier to determine if it is shorted.

Be Advised, a **definite pattern** of darkened commutator segments (every third or fourth segment, for instance), is often mistakenly assumed to indicate a problem.

The design of the armature can produce patterns in the film that the brushes put down. Some characteristics of armature design that can cause current fluctuations and *bar patterning* are:

Odd turns in armature coils
Commutator bar and armature core slot combinations

The bottom line is, if the discoloration pattern is repeated around the entire diameter of the commutator, there is probably NOT a problem with the armature winding.

NEWLY INSTALLED MOTORS

The troubleshooting procedures outlined previously all apply to motors that fail after having been *in operation for* sometime. Now we will discuss troubleshooting motors that fail shortly after installation.

PROBLEM: NEWLY INSTALLED MOTOR DOES NOT START OR FAILS SHORTLY AFTER INSTALLATION

If a new motor or *newly repaired* motor malfunctions the first time it is put in service:

- Make sure the control unit is properly matched to the motors nameplate ratings.

- Determine if the overcurrent devices are properly sized and properly adjusted.

- Check the control unit's input and output connections.

- Check the motor lead connections to be sure they are *correct* and tight.

- Make sure the controller is functioning properly.

If these checks indicate no reason for the malfunction, gather all the facts that you can and call Electrical Equipment Company for assistance.

Some of the procedures outlined below require special equipment and great familiarity with the construction of DC motors. We highly recommend that they should be undertaken only with the aid and assistance of motor shop personnel. You will especially want the motor shop personnel there to protect your interests if you feel there are any manufacturer or repair warranty considerations.

PROBLEM: NEWLY INSTALLED MOTOR RUNS AT HIGHER THAN RATED RPM

Make sure the shunt field is properly connected for the voltage provided by the controller.

- If a *dual voltage* shunt field motor has the fields connected for *high* voltage and *low* voltage is applied to the field, the motor will run at higher than rated RPM.

- This condition will also produce high armature currents.

- To restore the speed of the motor to its normal range, reconnect the shunt field for **low** voltage.

A note of caution: A shunt field connected (in parallel) for low voltage and then connected to a high voltage field supply will result in the shunt field burning out if adequate overcurrent protection is not provided. The manufacturer or the repair shop could NOT warrant such an overcurrent condition.

Reversed Series Field polarities may cause compound-wound motors to run at *above* the nameplate RPM when the motor is under load.

- If the motor rotation is correct, go ahead and interchange the series field leads (S1 and S2).

An incorrectly rewound armature can also cause a DC motor to run above its rated speed.

This will happen if fewer turns were used in the new armature winding.

Another winding error can be that the repairman put the first coil lead down into the wrong commutator bar. Hence all of the coil ends in the top of the commutator bars are in the wrong position. This error will generally double, or halve the motors speed depending on the motors original connection, and which error was made.

PROBLEM: NEWLY INSTALLED MOTOR RUNS IN REVERSE DIRECTION

After a DC motor has been repaired, you will sometimes find that the direction of rotation has been reversed.

Correct this problem by interchanging the armature leads A1 and A2.

PROBLEM: SPARKING UNDER THE BRUSHES IN NEWLY INSTALLED MOTORS

It is not uncommon to find sparking under the brushes in a newly installed motor. In most cases minor adjustments will correct the problem.

Brush holder alignment:

Check that the brush holders align all brushes with the commutator bars and that equal spacing between the brushes is maintained.

Make sure the brushes are in the neutral position.

In the neutral position, the line marking on the rocker ring must match the marking on the end bell. Make any adjustments necessary.

If you suspect that the marking on the rocker ring is incorrect, you can determine the neutral position by following this procedure:

1. Unlock the brush rigging so that you can shift it freely.
2. Disconnect the shunt field leads from the wiring
3. Connect an AC voltmeter to the shunts of brushes on *adjacent* brush holders.
4. Apply single-phase power (110 VAC) directly to the shunt field winding.
5. Shift the brush rigging left, and right, while observing the voltage indicated by the AC voltmeter. The brushes are on neutral when minimum voltage is indicated (usually less than 1 volt AC).
6. Lock the brush rigging securely and mark the rocker ring and the endbell to indicate the correct neutral position.

Relative polarity of main poles and interpoles:

Check the polarity of all main poles and interpoles using a magnetic compass. The polarities are correct when the polarity of the interpole *is the same as that of the main pole preceding it in the direction of rotation.*

Should the polarities not be in the proper relationship, interchange the two brush holder leads.

Brush grade:

Check to be sure that the brush grade is compatible with the environment in which the motor operates. All brushes must be of the same grade.

Brush grade selection requires specialized knowledge of the carbon materials and fabrication methods used. Electrical Equipment Company can help you acquire the data needed to make informed decisions on specifying new brushes.

If the cause of the motor failure cannot be determined using procedures described in this *Tech Note*, the motor should be brought to the Electrical Equipment Company location nearest you. We will have the equipment and “know-how” that you need.

3 Phase Alternating Current Motor Troubleshooting

Based on the EASA TechNote “Troubleshooting DC Motors”

THIS IS A BRIEF COURSE INTENDED TO ACQUAINT
YOU WITH BASIC ELECTRIC MOTOR
TROUBLESHOOTING AND TESTING

CAUTION

If you have not been trained in how to work safely near live electrical circuits, do not attempt to measure line voltages. Find someone who has been trained in electrical safety and let him or her take voltage readings. Great care is needed to eliminate the possibility of DEATH or serious injury.

ALWAYS disconnect the power and verify all parts are dead

before touching or handling any parts of electrical equipment.

Lock out and tag out all electrical circuits.

Test for voltage before touching any components.

Check for and eliminate the danger of “stored energy” caused by raised or spring-loaded equipment.

The basic test equipment you will need to troubleshoot AC motors includes:

AC voltmeter

AC clamp-on ammeter

Ohmmeter

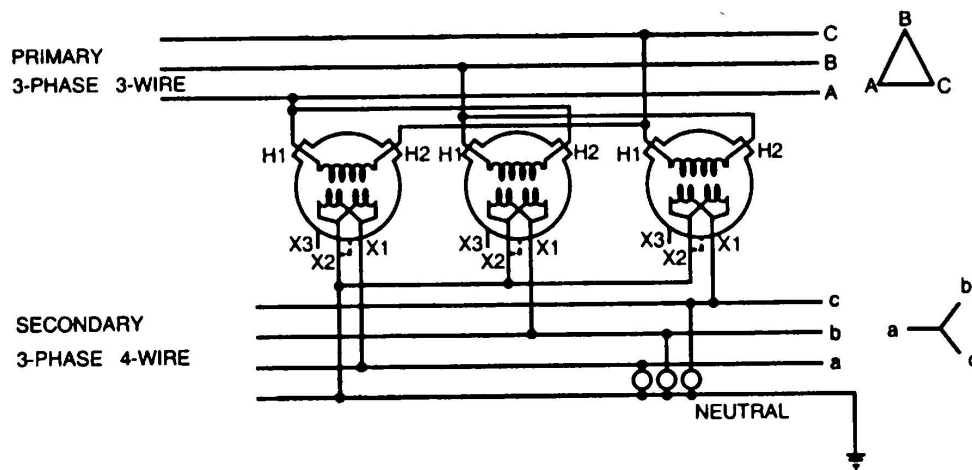
Megohmmeter

Voltage Tests

Voltage is the term used to describe the magnitude of the Electro-Motive Force, or in other words, the *pressure* at which electrons are being forced through a circuit.

It's current that kills, but it's voltage that really establishes the level of danger involved in working with electricity. Knowing the voltage you are working with enables you to take appropriate steps to safeguard yourself and those working near you from electrocution.

If you have not been trained in how to work safely near live electrical circuits, do not attempt to measure line voltages. Find someone who has been trained in electrical safety and let him or her take voltage readings. Great care is needed to eliminate the possibility of DEATH or serious injury.



Typical Delta/Wye Transformer Connections

Motors run while connected to the Secondary windings of a transformer bank. The transformers design and interconnection determines what voltage will be applied to your motors, as well as what voltage will be present from each line conductor to earth ground. (see a,b,c, neutral above)

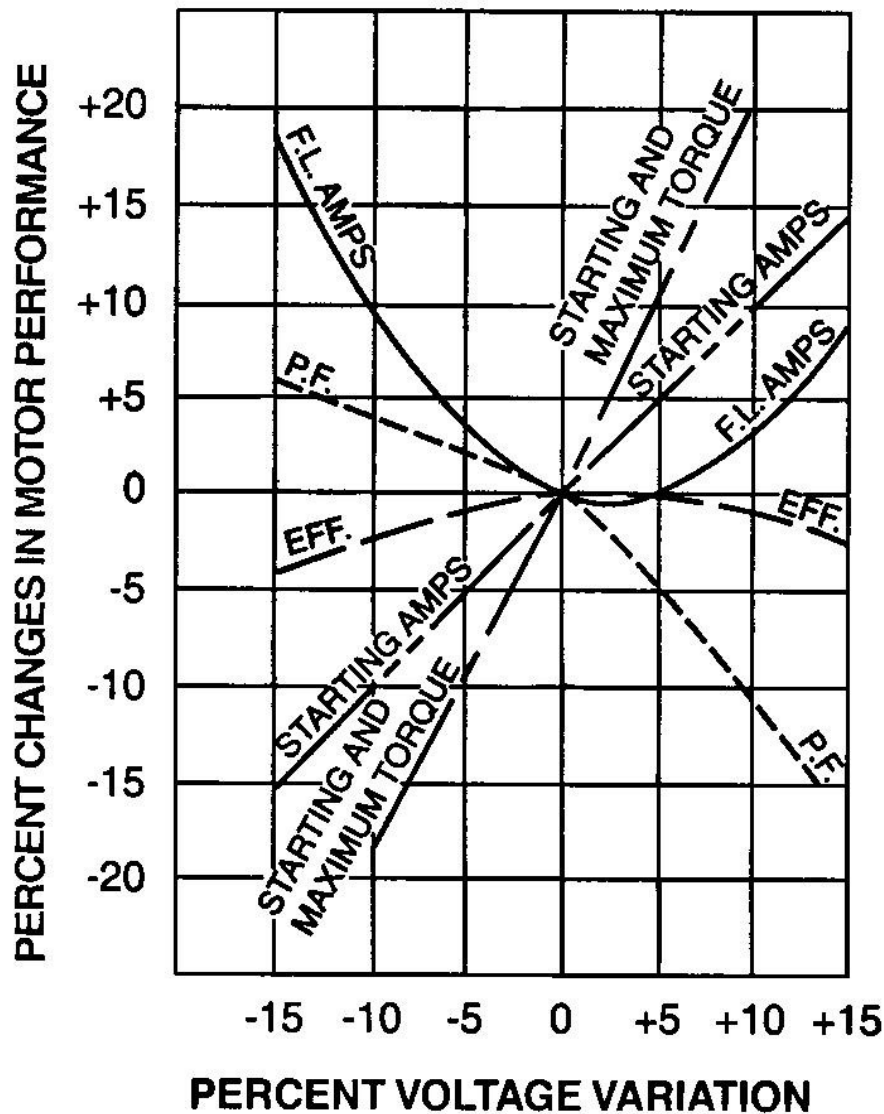
In Industrial plants today, the predominant voltage is 480 volts, Three-phase, sixty-cycles. Most motors are rated at 460 volts.

The voltage applied to your motors should not vary more than ten percent (plus or minus) from the motors rated voltage. That means a motor rated for 460 volts should have voltage applied that is between 414 and 506 volts. While motors will operate to their rated capacity at the lower end of

the voltage tolerance, their performance and overload capacity will be much better at the higher end of the range. Higher voltage is generally better for performance and less troublesome than lower voltage.

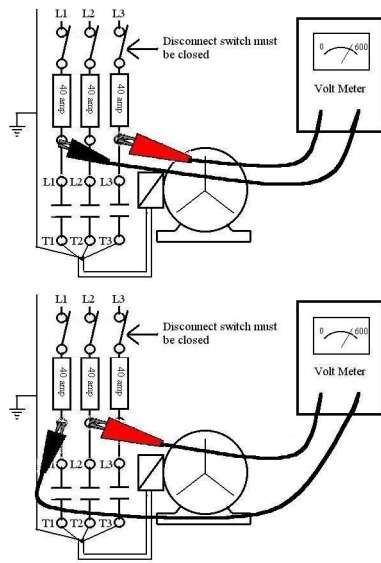
Effects of Voltage Unbalance

If the applied voltages are unbalanced, the motor in question may need to be **de-rated**. Voltage imbalance that is more than five percent of the line-to-line voltage will greatly reduce a motor's mechanical output and dramatically increase its internal heating.

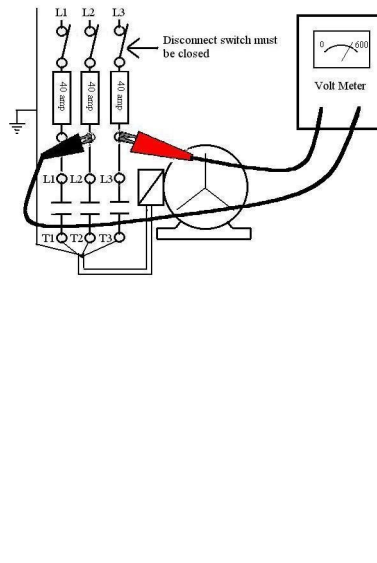


The graph above shows how *bad* things start to happen when the line-to-line voltages are unbalanced beyond 3 to 5 percent.

Basic voltage tests to identify applied voltage (motor is not running)



Line 1 to Line 3
2

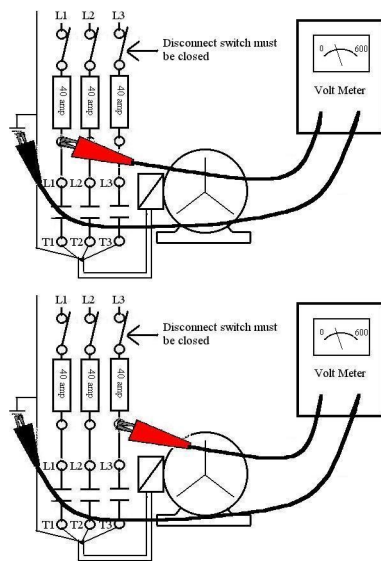


Line 2 to Line 3

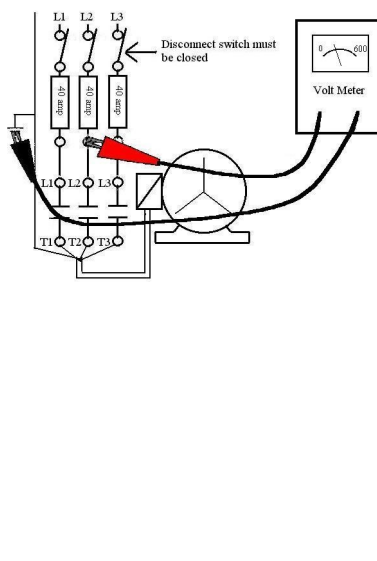
Line 1 to Line

In the process of checking for the presence, and *balance*, of all three-phase voltages, you may, by process of elimination find a blown fuse. The line that always reads “low volts” is the one with the blown fuse.

Voltage tests to verify “Line to ground” potentials and to isolate a blown fuse.



Line 1 to Grd.



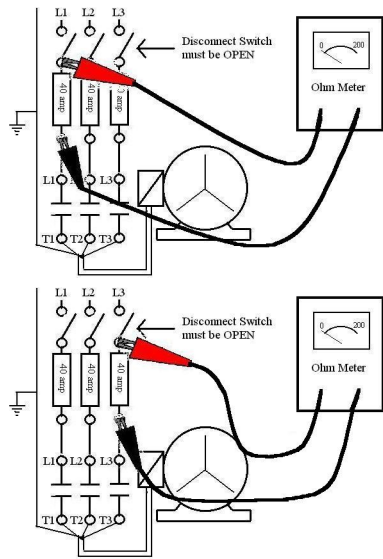
Line 2 to Grd.

Line 3 to Grd.

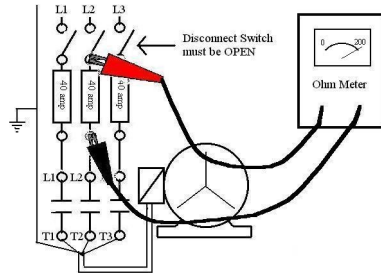
The blown fuse should read only a few “milli-volts” to earth ground. The good fuses should read normal line to ground potentials.

Continuity test to confirm blown fuse

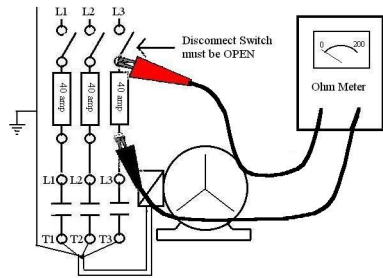
Be aware that in the event of a heavy fault current, “carbon tracking” can occur within the blown fuse and produce a volt reading that can confuse a very sensitive voltmeter *and* you. So a final “Continuity Test” should be performed. **Be certain to pull the disconnect to its OFF position before doing your continuity test. Be sure to repeat your first series of tests on the TOP END of all three fuses to verify that the power is off.**



Test fuse 1



Test fuse 2



Test fuse 3

Any blown fuse will read a high resistance.

Ground Fault Tests

AC motor windings are NOT to be grounded.

There are to be no electrical connections from electrical windings to earth ground.

(Exception: alternators, some transformer windings)

The unit of measurement for electrical resistance is the ohm (Ω)

Electrical Resistance is a numeric value assigned to the relative inability of materials to transfer electrons from one molecule to the next.

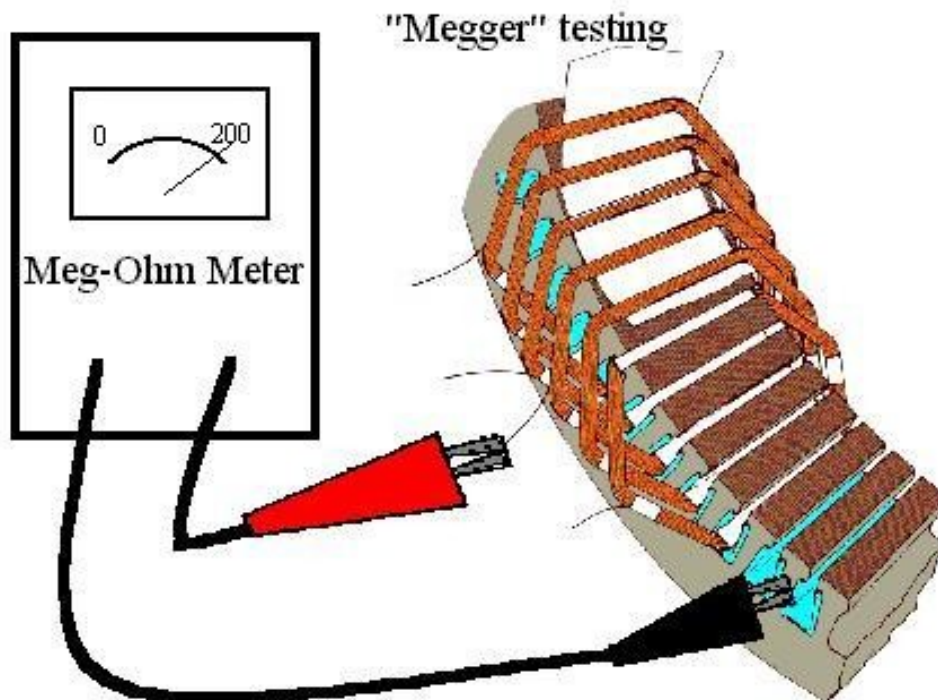
One Ohm is the amount of resistance that lets 1 Volt make 1 Amp of current to flow in a conductor.

One Meg-Ohm equals 1,000,000 ohms (high resistance)

One Milli-Ohm equals $1/1000$ ohm (low resistance)

All windings, whether connected to earth ground or not have **“Ground Wall Insulation”**.

Ground Wall Insulation keeps the electricity from getting to earth ground in the wrong place. If electricity gets to earth ground too soon, it doesn't do the work we want it to do.



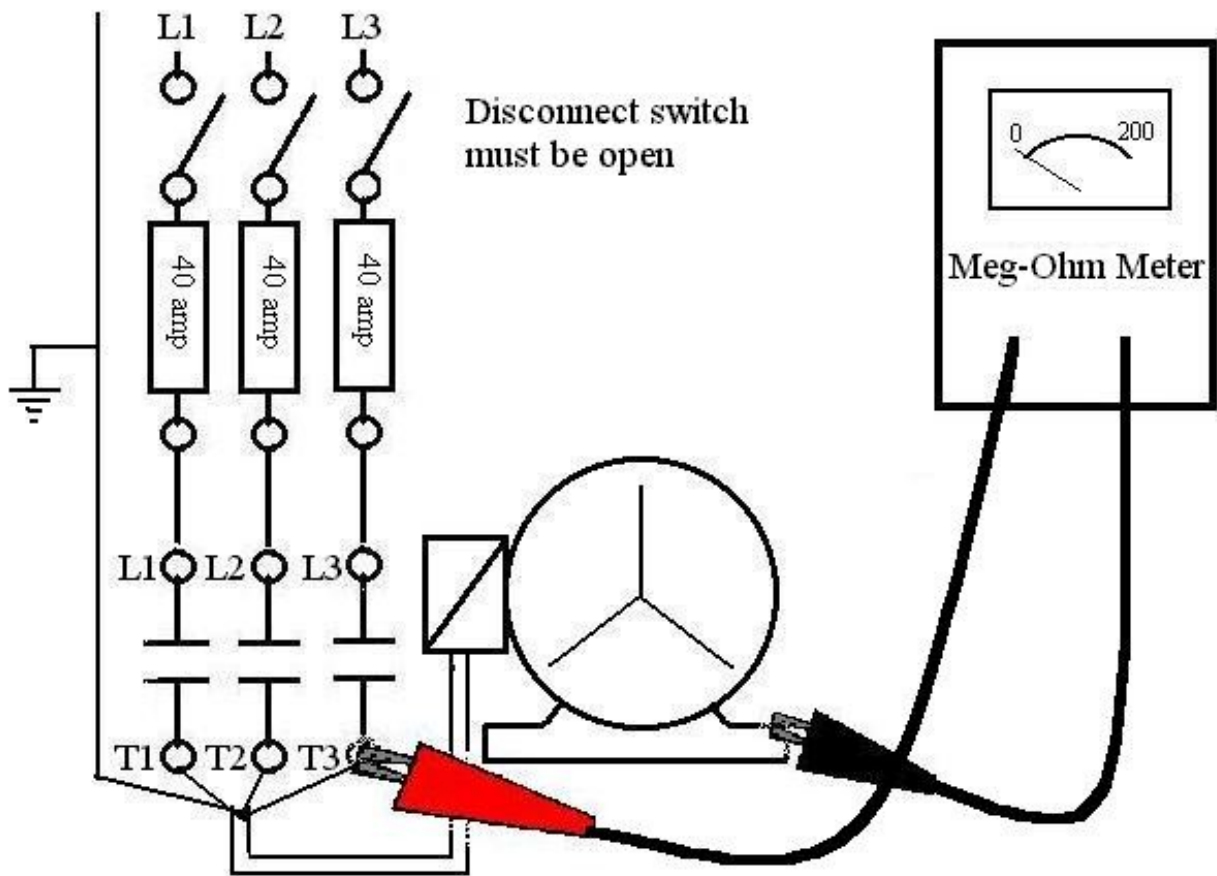
Your “Megger Testing” is to verify that no damage has been done to the “Ground Wall Insulation”. (Ref: Ground Wall Insulation is the Blue insulation in the figure above)

A Meg-Ohm meter will use a High Voltage Potential (usually 500 or 1000 Volts) to “Push” or “Stress” the limits of electrical insulation. The high voltage is required in order to give you a meaningful measurement of the High Resistance. (Meg-Ohms, Millions of Ohms) that should exist across

the “Ground Wall”. A Meg-Ohm Meter is used to find “**failures**” in electrical insulation.

When using a Meg-Ohm Meter you connect one lead to the winding, and the other lead to the frame of the unit under test. When you activate the Meg-Ohm Meter you are impressing 500, or 1000 volts of pressure against the “Ground Wall Insulation”. You are trying to *force* electrons to get through the Ground Wall Insulation.

Megger Testing an installed motor



If your motor is connected to an “electronic drive”, disconnect the wiring from the drive terminals before doing your megger testing.

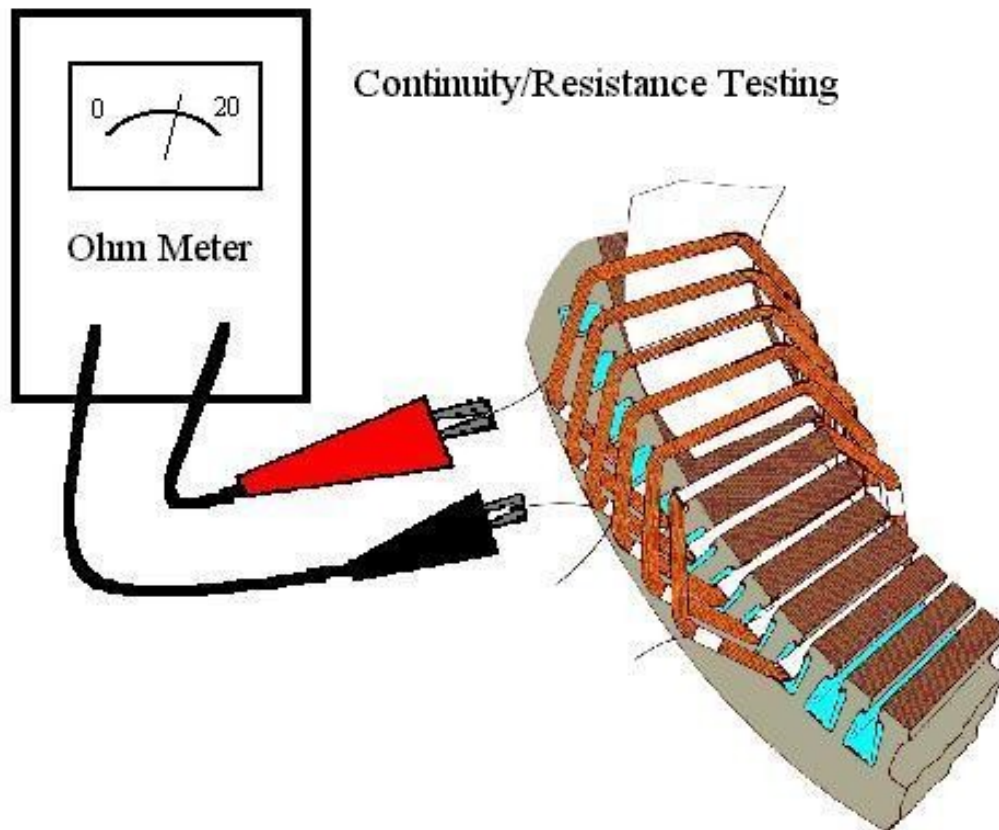
A winding can burn off, or “open” when a large fault occurs. Be sure to check all three lines to the motor before saying the motor and wiring is OK.

POP QUIZ!

Is the motor under test above GOOD or BAD?

Continuity Tests

You can use an Ohm Meter to find what wires are connected to specific circuits. In the process you can determine the resistance of the circuit in “Ohms” and make comparisons of equivalent circuits.



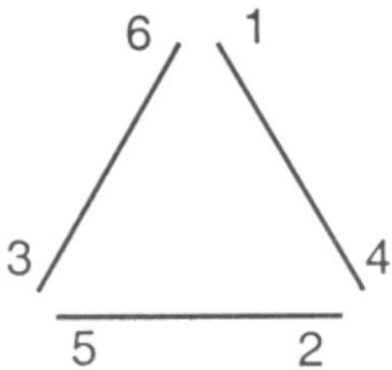
In the example above the Ohm Meter is being used to measure the resistance on a single coil group.

An Ohmmeter uses a Low Voltage Potential, (Usually 1 to 3 volts) to measure electrical resistance or check “continuity”.

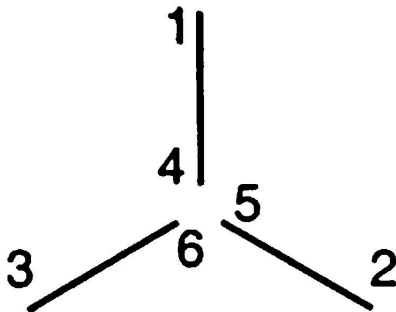
Every motor has distinct coil groups that are connected internally in the motor to comprise the phase windings. In troubleshooting a motor you may need to verify that the motor lead numbers are correct, and that there have been no electrical faults that create “short circuits” between the different phases.

Every good electrician knows the lead numbering sequence of three phase motors, or he has diagrams available for ready reference. The EASA Electrical Engineering Pocket Handbook was specifically designed for this purpose.

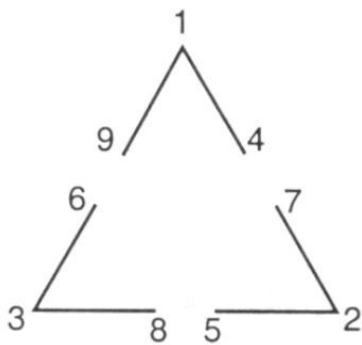
Here are some examples of motor lead numbering systems. Each arrangement has its own special application.



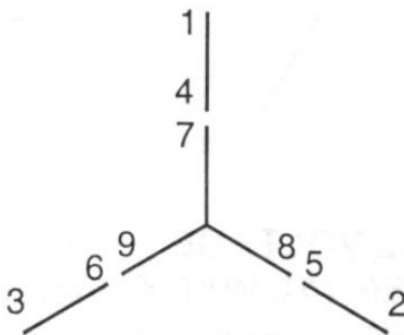
Six Lead Delta



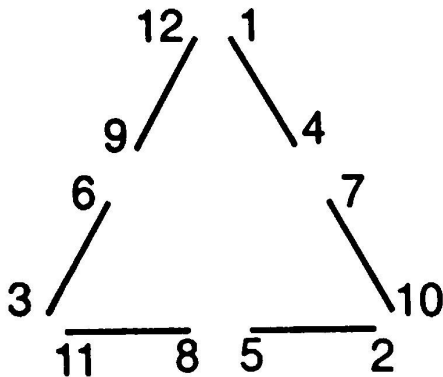
Six Lead Wye



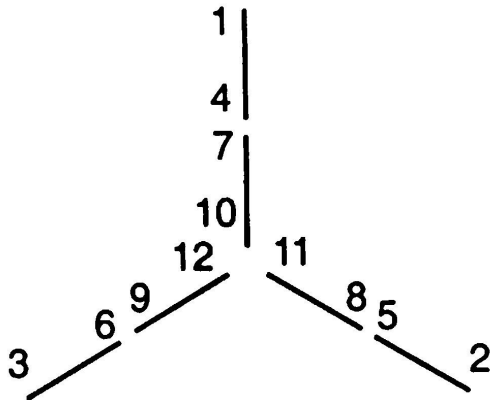
Nine Lead Delta



Nine Lead Wye

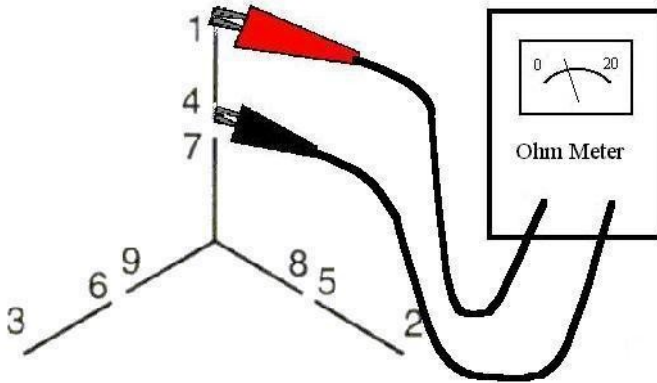


Twelve Lead Delta



Twelve Lead Wye

Continuity Testing of Motor Windings

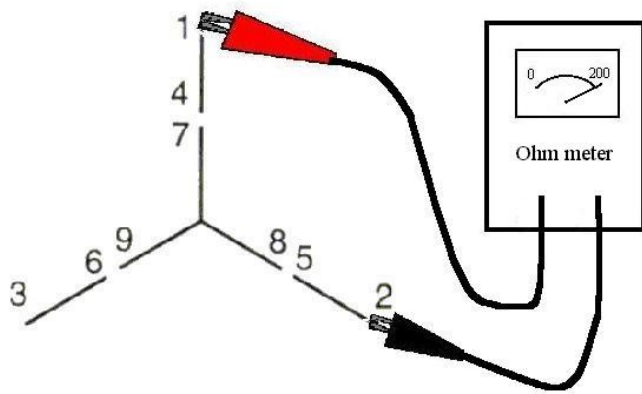


The line between #1 and #4 represents a circuit in the motor.

The ohmmeter should show continuity when connected to #1 and #4 because they are the opposite ends of a circuit in the motor. Your ohmmeter will give you a reading.

In this example the ohmmeter is connected to different sections of the winding, where no connection should exist.

If the winding is OK, in this instance, the ohmmeter should indicate a high resistance because there is no circuit.



By using motor connection diagrams as a reference you can make tests to different combinations of leads and determine if the internal circuits in the motor are defective.

Any defects in the winding indicate that the motor will need to be removed from service and evaluated for repair or replacement.

TESTING MOTORS WITH A PREVIOUS HISTORY OF SUCCESSFUL OPERATION

If the motor has been operated successfully, problems such as incorrect hook-up or internal misconnection can be ruled out immediately.

Before proceeding,

Read and record pertinent motor nameplate data.

HP

RPM.

Rated voltage

Rated current

Frame size

Enclosure

Look over the installation and inspect the motor for any obvious defects that would prevent safe operation and testing. Look for:

Damaged windings

As evidenced by smoke deposits or copper particles in J-box

Loose connections in J-box (melted wire nuts, burned insulation, arcing to cover or box)

Broken or missing parts (Pulleys, belts, covers, etc.)

PROBLEM: MOTOR WILL NOT START

Check to make sure all three phases are present at the control unit. (Use AC volt meter)

Three phase motors will not start on single phase current.

If the main fuse is blown, DO NOT apply power to the motor until you have replaced defective fuses and checked for any ground faults in the motor and its wiring.

Check for ground faults:

Disconnect the motor's power source. (Open the disconnect switch and verify with your voltmeter that the power has been disconnected "downstream" of the switch)

Use the megohmmeter to measure the insulation resistance of all windings to earth ground.

Take care to isolate the motor from any "electronic controls" such as soft starters and frequency drives before using the meg-ohm meter. You may have to undo the motor leads at the controls terminals before testing. The voltages from a Meg-ohm meter could possibly damage the controls.

Any "grounded" conditions must be corrected before power is applied to the motor.

Check to see that the motor will turn over by hand. Remove any obstructions or free up the jammed machine if that condition exists. Find out now if the motor bearings are rough or wiped out.

Inspect Motor Connections

Inspect electrical connections to the motor in the control and in the motor's J-box.

Correct any loose or broken connections.

Check for signs of heating or "resistive connections"

If the main fuses are OK, all ground faults have been removed, and the machine will rotate by hand, prepare to attempt a restart.

Attempt a restart:

Position yourself away from rotating equipment, with the motor remaining in your sight. If necessary, get help initiating the start signal, so you can observe the motor during start-up. Instruct your helper so he is prepared to quickly shut down the motor at your signal if a problem develops.

Set your (digital) clamp-on ammeter to its highest range and attach it to one of the lines feeding the motor. Be aware of what the motor's full load amp rating is.

(Be careful if you are using an analog ammeter. High inrush currents could damage the meter movement)

Close the disconnect switch, and start the motor.

Watch for rotation to begin, being careful to immediately disconnect the motor if it fails to rotate.

If the motor fails to rotate when the power is applied, disconnect the power and resume testing to determine the problem.

As the motor accelerates, observe rotation, and listen to the sound of the motor. Remain prepared to quickly shut the motor off if it does not continue to accelerate smoothly to full speed. Be careful to notice if the motor “hangs” at a fixed speed and fails to finish its acceleration. If the acceleration to full speed does not occur smoothly, immediately shut down the motor and proceed with other testing.

While the motor is accelerating, check your ammeter so you can observe the starting currents diminish as it reaches full speed.

When the amps fall off to normal operating levels, quickly move your ammeter to each Line in order to check all three phases. Verify that the motor currents are “even”, and that they do NOT exceed the motor's rated amperage. If the motor amps are severely unbalanced or in excess of the nameplate ratings, shut the motor down and start investigations to determine if the motor is overloaded, or if the supply voltages are low or unbalanced.

In the event of unbalanced currents, check the applied voltage as near to the fully loaded motor as is safe, to verify that the applied voltages are even. Motor voltage unbalance should not exceed 5% of line voltage. For a 460 volt motor, that is 23 volts variance line to line. If you cannot read the voltage close to the motor, consider the length of the run and size of wire to get a grip on actual voltage drop at the motor.

Any voltage unbalance will significantly reduce the output capacity of a motor. Current imbalance over the 5% range dictates that the motor's load be reduced to compensate for the lost power.

If the line voltages are **even** and the **current imbalance still exceeds 10%**, the winding is probably shorted and the motor should be repaired.

In the event of a motor running overcurrent, disconnect the load and restart the motor. With the motor running unloaded, verify that the “No Load” currents are within the following guidelines.

900 – 1200 rpm motors
70% Full Load amps

Approximately 50 to

1800 rpm motors

Full Load amps

3600 rpm motors

30% Full Load amps

(Some may be higher)

Approximately 30%

Approximately 20 to

If the No-Load currents are reasonably balanced, and within the suggested limits, the motor is probably being overloaded. Reduce the load or install a larger motor.

If the uncoupled motor's No-Load currents significantly exceed the above guidelines, or the currents are grossly uneven, it is safe to assume that the windings are shorted and the motor is in need of repair. A shorted winding will also produce a "labored", "whining" sound that is quickly identifiable to the experienced ear. Of course, watch for smoke.....

Test and inspect controller (Soft start, Adjustable Frequency Drive)

If no output is read from the controller, determine if the problem is in the control circuit and correct it.

Is the controller tripped? Modern Variable Frequency Drives have some pretty sophisticated troubleshooting aids.

If the VFD has "faulted", proceed to determine the cause and correct the problem.

Over current (Excessive load over a period of time)

Over voltage (Overhauling type of load)

Over Heat (High ambient temperatures-Overloading)

Are the thermostats in the motor tripped (N/C contacts)?

Attempt a reset.

Make sure the controller is getting a start signal (N/O contacts)

Make sure there is not a STOP signal (N/C contacts)

If the controller isn't functioning by this point, it's pretty safe to say that the controller is defective.

PROBLEM: OVERLOAD RELAY TRIPS; OR FUSES BLOW WHEN MOTOR STARTS

A starting current that is too high, or lasts too long, will cause tripping of the overload relay or blow fuses. Motor starting currents that don't diminish quickly

will be too high to be sustained by normal overload protection. The motor and its associated load must accelerate quickly. If acceleration is delayed due to increased load nuisance, tripping can be the result.

Grounded windings.

Test all windings for ground failure using the megohmmeter. Any grounded windings must be repaired before power is applied to the motor

Mechanical problems with the motor or driven equipment.

Mechanical problems such as worn bearings or other problems with the motor or machine could cause a mechanical overload. Determine if the problem is in the motor itself or in the driven equipment. Uncouple the motor and turn the rotor by hand. Check for bad bearings or other mechanical binding.

Shorted windings

If the rotor turns freely, attempt a restart as outlined earlier, and check the no load currents in comparison to the amperage guidelines stated earlier. If the motor starts and runs within those limits, the problem is most likely in the driven equipment and not in the motor.

PROBLEM: MOTOR RUNS AT LOWER THAN RATED RPM

AC squirrel cage motors run at a continuous speed, unless they are a special multi-speed design, or if they are connected to a Variable Frequency Drive.

If you have a normal motor installation, and the speed of the load varies, check your motor currents to see that the motor isn't being overloaded.

In most cases you will find the motor is running as it should, but slipping belts or other mechanical problems are letting the *load* vary in speed.

Rotor testing

There is however an instance where the motor currents don't seem excessive, the belts are tight enough, but the motor doesn't seem able to pull the load. These are RARE instances, but if these are the facts, then you can suspect a bad rotor.

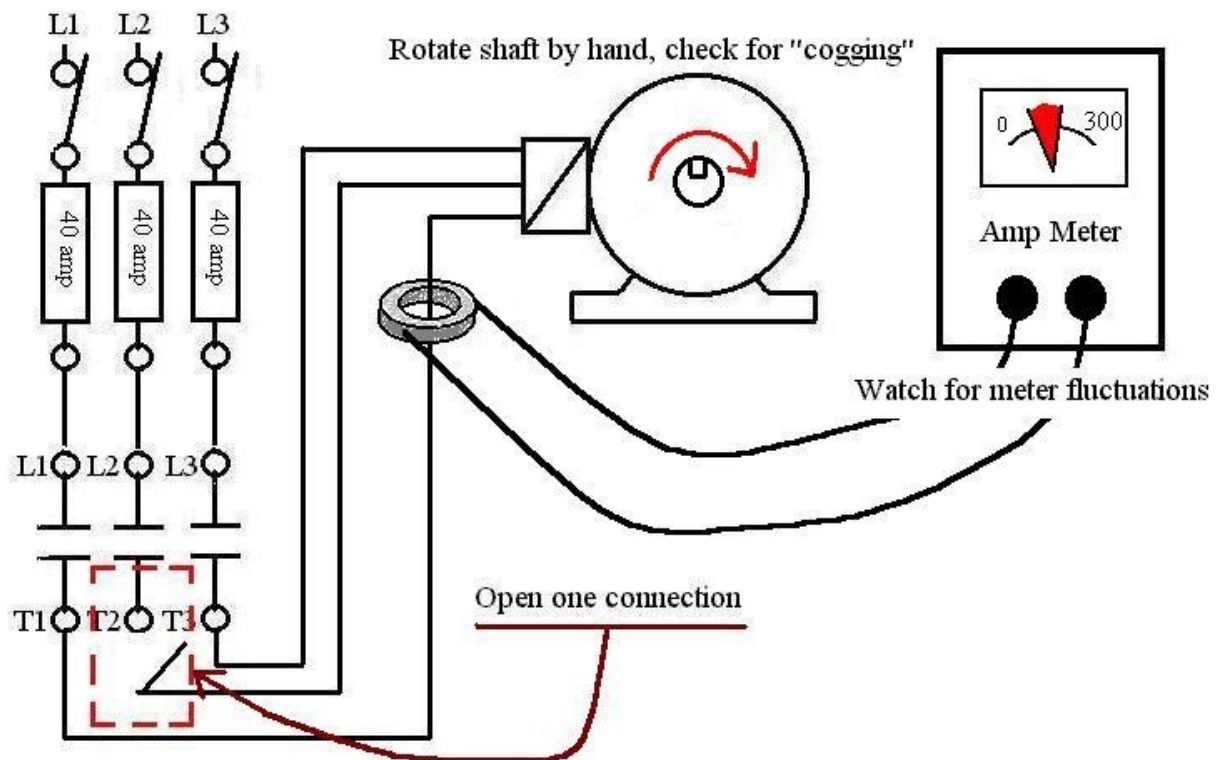
Broken rotor bars will greatly reduce a motor's torque and still allow the currents to remain "reasonable", *if* the motor is not severely overloaded. The testing described here requires thorough preparation and assistance from another mechanic or electrician.

The motor winding can be "single-phased" to test the rotor. That is to say that you will disconnect one phase of the motor winding, and energize the remaining two phases. Under these conditions, motors with broken rotor bars will exhibit a "cogging" effect while the shaft is being rotated by hand. The current being applied to the stator winding will also fluctuate correspondingly to the rotor's "cogging".

This test will produce potentially damaging currents, so it must be conducted quickly and with great care for your personal safety.

This test should NOT be conducted using line voltages on motors greater than 100hp.

SINGLE PHASE ROTOR TEST



1. With the power disconnected, open one phase at either the motor starter, or the motor's J-box.
2. Disconnect the motor from its load. (Remove belts/open coupling)
3. Attach an AC Ammeter to one of the *connected* motor leads.
4. Set the ammeter to a scale that is 200 to 300 percent of the motor's full load current.
5. Close the Disconnect Switch.
6. At your direction, have your assistant apply power to the motor. Immediately rotate the motor shaft by hand while feeling for a pronounced "cogging" effect. While doing so, you or your assistant should observe the ammeter for deflections in its reading.
7. Shut off the power. The entire test sequence should be accomplished in less than ten seconds to avoid blowing fuses or damaging the motors windings.

If the shaft turns freely, with very little movement of the ammeter, you can conclude that the rotor is OK.

If you find cogging and variable motor currents, the rotor has open bars requiring the motor to be repaired or replaced.

NEWLY INSTALLED MOTORS

The troubleshooting procedures outlined previously all apply to motors that develop problems after having been *in operation for* sometime. Now we will discuss troubleshooting motors that give problems during or shortly after installation.

PROBLEM: NEWLY INSTALLED MOTOR DOES NOT START AFTER INSTALLATION

If a new motor or *newly repaired* motor malfunctions the first time it is put in service:

Check the control unit's input and output connections.

Are the incoming line connections made at the correct points in the controller?

Are the output connections made at the correct points?

Are all the connections tight and secure?

Make sure that all three phases are present at the input of the motor controller.

Measure the Line voltages to verify that they are present and evenly balanced.

Make sure the supply voltages are correct.

Check the motor nameplate to verify that the line voltages agree with the nameplate rating of the motor.

Check the motor lead connections to be sure they are *correct* and tight.

Inspect the line and motor lead connections in the motor's J-box.

Are the connections tight and well insulated?

Are the line connections made to the appropriate motor leads?

Are all of the motor leads securely and properly connected?

Make sure the controller is functioning properly.

In the case of an electro-mechanical motor starter, does the contactor close securely?

In the case of a Variable Frequency Drive, does output result on the initiation of a start signal?

Determine if the overcurrent devices are properly sized and properly adjusted.

Is the overload tripped?

Is the overload correctly sized for the motor?

In the case of an Adjustable Frequency Drive, Is the drive faulted?

PROBLEM: NEWLY INSTALLED MOTOR RUNS IN REVERSE DIRECTION

To reverse the rotation of a three-phase motor, switch any two incoming lines.

Swapping line connections is the simplest option, but in the case of large motors where the incoming lines are too large and difficult to move easily, careful study

may be needed to decide how to rearrange the motor leads inside the controller. Special reduced voltage starting arrangements complicate reconnection.

If you have more than three motor lead conductors connected to your motor starter, call your friends at Electrical Equipment Company for assistance. We'll be glad to help!

